

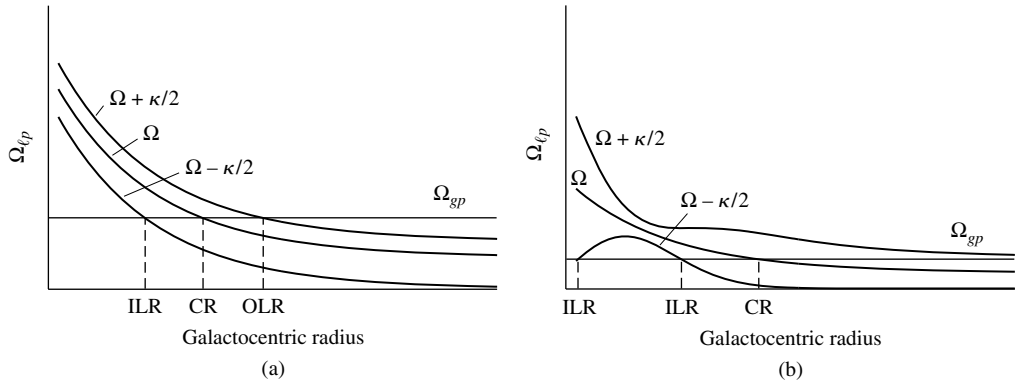


FIGURE 25.30 The existence of resonance radii depends on the global angular pattern speed of the arms and the shape of the galaxy's rotation curve. (a) A galaxy with a single inner Lindblad resonance (ILR), a corotation resonance (CR), and an outer Lindblad resonance (OLR). (b) A galaxy with two ILRs, a CR, and no OLR. Note that for sufficiently large values of Ω_{gp} , there may not be any ILRs.

continually building the waves back up again. This situation is analogous to that of stellar pulsation (Chapter 14); regions of partial ionization (primarily hydrogen and helium) drive pulsations, while other regions of the star damp the oscillations.

Nonlinear Effects in Density Wave Theory

Despite a great deal of work, a complete understanding of density waves has not yet been fully realized. For instance, various effects not considered in the simple model presented here may play important roles, such as nonlinear (higher-order) terms in Φ_{eff} (recall Eq. 25.28). Also, the waves themselves alter the gravitational potential in which they originate so that azimuthal symmetry breaks down.

One important driving mechanism in a number of grand-design spirals, including M51 (Fig. 25.17), is the presence of a companion galaxy that triggers spiral structure through tidal interactions (see Section 26.1 and Fig. 26.10). A more detailed discussion of the physical driving mechanisms that help to establish and maintain density waves is beyond the level of the current discussion, but it is worth pointing out that the techniques for studying spiral density waves have much in common with the theoretical procedures used to investigate stellar pulsation; both linear and nonlinear models have been employed in the investigation of spiral structure.

Numerical *N*-Body Simulations

An important example of a nonlinear *N*-body simulation of a rotating disk is one that was calculated in an early work by F. Hohl in 1971 (see Fig. 25.31). In that calculation, 10^5 stars were initially placed in axisymmetric orbits ($t = 0$), but as the simulation progressed, Hohl's model proved to be very unstable against the development of the $m = 2$ pattern (also known as the $m = 2$ mode), and a two-armed spiral density wave developed ($t = 2.0$). As the simulation continued, the disk became "hot," meaning that the velocity dispersion of

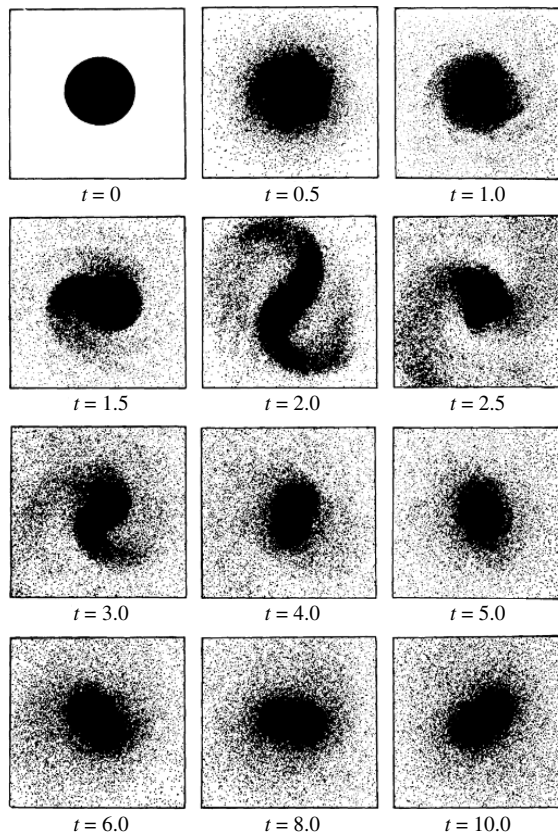


FIGURE 25.31 An early study of a rotating disk with an N -body simulation using 10^5 stars. The disk began with complete axial symmetry and quickly developed an $m = 2$ mode instability. Eventually the disk “heated up,” destroying the spiral arms but leaving a long-lived bar. (Figure from Hohl, *Ap. J.*, 168, 343, 1971.)

its stars became large relative to their orbital velocities, and the spiral structure dissipated. Interestingly, a bar instability persisted throughout the rest of the simulation, and the final structure bears a strong resemblance to an SB0 galaxy.

Such bar instabilities have proved themselves to be common features in N -body calculations. Apparently a rotating disk is highly susceptible to a bar-mode instability, at least on computers. Of course, there is strong support for the idea that bar modes are favored in real galaxies as well, since roughly two-thirds of all disk galaxies do exhibit bar-like structures in their centers.

Hohl’s work does suggest one possible means of stabilizing disks against various mode instabilities: the presence of high-velocity dispersions of its stars. In his original simulations, Hohl’s bar heated the disk, destroying the $m = 2$ trailing-arm mode. As we will see in Section 26.1, tidal interactions and/or mergers may also play important roles in heating the disks of spiral galaxies.

In modern N -body simulations, models have also included the effects of highly centralized masses, such as the gravitational influence of supermassive black holes. Various researchers have shown that the inclusion of sufficiently condensed and massive central mass concentrations can weaken, and even destroy, the bar instability. Most studies find that a mass on the order of a few percent of the mass of the galaxy's disk is required in order to affect the bar.

Stochastic, Self-Propagating Star Formation

Although most of the discussion found in this section has focused on the Lin–Shu hypothesis of quasistatic density waves and the existence of grand-design spirals, many of the galaxies observed in the universe are flocculent spirals. It may be that these objects are composed of a linear combination of several stable density perturbations (somewhat reminiscent of modes in stellar oscillations), along with sufficient patchiness in the interstellar medium to give the appearance of less well-defined spirals. Alternatively, it may be that an entirely different mechanism is responsible for the spiral structure seen in these galaxies.

In 1976 M. W. Mueller and W. David Arnett proposed a theory of spiral structure for flocculent spirals known as **stochastic, self-propagating star formation (SSPSF)**. In their theory they imagine spiral structure arising from outbursts of star formation that propagate across the galaxy. When one region of the galaxy undergoes a star formation episode, its most massive stars will age rapidly, producing core-collapse supernovae. The supernovae shock waves will travel through the interstellar medium, triggering the collapse of other gas clouds in nearby regions, where further star formation will occur, allowing the process to continue. This scheme has been compared to a forest fire, with the flames jumping from tree to tree. Spiral structure arises when the differential rotation of the galaxy draws these newly “lit” regions into trailing arms.

Although SSPSF has been successful in producing flocculent spiral structure in computer simulations, it is unable to account for the transitions from dust lanes to OB stars to red stars across the spiral arms of grand-design spirals, so it probably cannot explain those galaxies. Of course, it may well be that we will need both of these theories (and perhaps others) to understand the abundance of spiral structure found throughout the universe.

25.4 ■ ELLIPTICAL GALAXIES

Although Hubble type correlates well with a wide variety of physical parameters for late-type galaxies, thereby ensuring its continued usefulness for those systems, the Hubble-type designation for early galaxies (which is based solely on apparent ellipticity) has shown itself to be virtually irrelevant in terms of trying to categorize other characteristics. As a result, subtype distinctions are made for ellipticals that are independent of the ellipticity, focusing instead on other morphological features, such as the size, absolute magnitude, and surface brightness of the image. Once thought to be the simplest of the major galaxy types, since the 1980s ellipticals have come to be seen as remarkably diverse and complex. As we shall see in Section 26.1, some of this complexity may arise, at least in part, from strong environmental evolution, possibly involving tidal interactions or mergers with neighboring galaxies.

Morphological Classes of Elliptical Galaxies

Today a number of separate morphological classes are commonly used to distinguish among the elliptical galaxies:

- **cD galaxies** are immense, but rare, bright objects that sometimes measure nearly 1 Mpc across and are usually found only near the centers of large, dense clusters of galaxies. Their absolute B magnitudes range from less than -22 mag to -25 mag, and they have masses of between 10^{13} and $10^{14} M_{\odot}$. cD galaxies are characterized by having central regions with high surface brightnesses ($\mu = 18$ B -mag arcsec $^{-2}$) and very extended, diffuse envelopes ($\mu = 26$ to 27 B -mag arcsec $^{-2}$). They may also possess tens of thousands of globular clusters, with typical specific frequencies, S_N , of 15 (see Eq. 25.15). Furthermore, these galaxies are known to have very high mass-to-light ratios, sometimes exceeding $750 M_{\odot}/L_{\odot}$, implying large quantities of dark matter.
- **Normal elliptical galaxies** are centrally condensed objects with relatively high central surface brightnesses. They include the giant ellipticals (gE's), intermediate-luminosity ellipticals (E's), and compact ellipticals (cE's). The absolute B magnitudes of normal E's range from -15 to -23 , masses of between 10^8 and $10^{13} M_{\odot}$, diameters from less than 1 kpc to nearly 200 kpc, mass-to-light ratios from 7 to more than $100 M_{\odot}/L_{\odot}$, and specific frequencies of globular clusters in the range of 1 to 10. Lenticular galaxies (S0's and SB0's) are often grouped with normal E's.
- **Dwarf elliptical galaxies** (dE's) have surface brightnesses that tend to be much lower than those of cE's of the same absolute magnitude. The absolute B magnitudes of dE's fall between -13 and -19 , they have typical masses of 10^7 to $10^9 M_{\odot}$, and their diameters are on the order of 1 to 10 kpc. Their metallicities also tend to be lower than for normal E's. The average value of the specific frequency of globular clusters is $\langle S_N \rangle = 4.8 \pm 1.0$, which is still higher than for spirals.
- **Dwarf spheroidal galaxies** (dSph's) are extremely low-luminosity, low-surface-brightness objects that have been detected only in the vicinity of the Milky Way. Their absolute B magnitudes are only -8 to -15 mag, their masses are roughly 10^7 to $10^8 M_{\odot}$, and their diameters are between 0.1 and 0.5 kpc.
- **Blue compact dwarf galaxies** (BCD's) are small galaxies that are unusually blue, with color indices ranging from $(B - V) = 0.0$ to 0.3 . This corresponds to main-sequence stars of spectral class A, indicating that these galaxies are undergoing particularly vigorous star formation. They have absolute B magnitudes of -14 to -17 , masses on the order of $10^9 M_{\odot}$, and diameters of less than 3 kpc. As is expected for very active star formation, BCD's also have a large abundance of gas, with $M_{\text{HI}} = 10^8 M_{\odot}$ and $M_{\text{H II}} = 10^6 M_{\odot}$ constituting roughly 15% to 20% of the entire mass of the galaxy. They also have correspondingly low mass-to-light ratios; in an extreme case, ESO400–G43 has $M/L_B = 0.1$ despite the dominance of dark matter at large radii.

TABLE 25.3 Characteristic Data for cD, Elliptical, and Lenticular Galaxies.

	cD	E	S0/SB0
M_B	−22 to −25	−15 to −23	−17 to −22
M ($M_\odot$)	10^{13} – 10^{14}	10^8 – 10^{13}	10^{10} – 10^{12}
Diameter (D_{25} , kpc)	300–1000	1–200	10–100
$\langle M/L_B \rangle$ ($M_\odot/L_\odot$)	> 100	10–100	~ 10
$\langle S_N \rangle$	~ 15	~ 5	~ 5

TABLE 25.4 Characteristic Data for Dwarf Elliptical, Dwarf Spheroidal, and Blue Compact Dwarf Galaxies.

	dE	dSph	BCD
M_B	−13 to −19	−8 to −15	−14 to −17
M ($M_\odot$)	10^7 – 10^9	10^7 – 10^8	~ 10^9
Diameter (D_{25} , kpc)	1–10	0.1–0.5	< 3
$\langle M/L_B \rangle$ ($M_\odot/L_\odot$)	~ 10	5–100	0.1–10
$\langle S_N \rangle$	4.8 ± 1.0	—	—

Tables 25.3 and 25.4 summarize some of the characteristics of early-type galaxies. Lenticular galaxies have also been included in the list, since they appear to have much in common with many ellipticals.

Surface Brightness Profiles

cD's and normal ellipticals have surface brightness profiles that closely follow the $r^{1/4}$ law (Eq. 25.2). As an example, the brightness profile of NGC 3379, a gE galaxy, is shown in Fig. 25.32. However, as the mass of the galaxy decreases, there is a fairly smooth transition over to an exponential profile (Eq. 25.3). This is particularly true of dE's and dSph's.

Dust and Gas in Elliptical Galaxies

The low gravitational binding energy of dE's and dSph's means that it would be very difficult for these galaxies to retain a significant amount of gas, which is indeed what is observed. In fact, dSph's are virtually devoid of gas. Consequently, dE's and dSph's are not actively forming stars today. Moreover, they have very low metallicities, with values similar to those found in globular clusters. It has been suggested that these systems lost most of their gas via supernova-driven mass loss or by **ram-pressure stripping** as the galaxies passed through the gas found in clusters of galaxies (see Section 27.3).

For many years it was believed that all elliptical galaxies had been largely stripped of any dust or gas that had not yet formed into stars or, alternatively, that star formation had proceeded very efficiently during the earliest history of these galaxies, depleting all available gas. However, we have now come to realize that gas and dust are present within most normal elliptical galaxies, albeit at somewhat diminished levels relative to spirals.

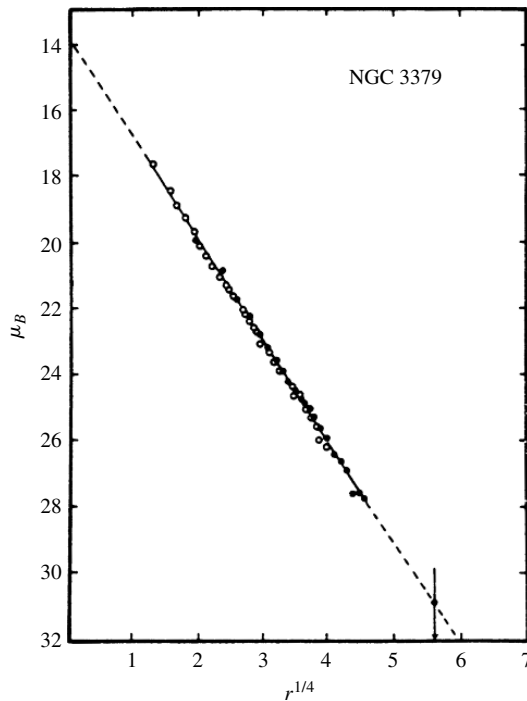


FIGURE 25.32 The surface brightness profile of the giant elliptical galaxy NGC 3379 is well represented by an $r^{1/4}$ law. (Figure adapted from de Vaucouleurs and Capaccioli, *Ap. J. Supp.*, 40, 699, 1979.)

The hydrogen gas that is present in E/S0's is usually evident in several forms. A 10^7 K X-ray-emitting hot-gas component of 10^8 to $10^{10} M_\odot$ makes up the majority of all the gas that is present in a typical E/S0 galaxy. (Possible heating mechanisms include supernovae and stellar winds.) A warm-gas component (10^4 to $10^5 M_\odot$ at 10^4 K), observable in $H\alpha$, is also present in the form of H II regions. Furthermore, 10^7 to $10^9 M_\odot$ of cold H I ($\sim 10^2$ K) is detectable at 21 cm. CO emission studies indicate that 10^7 to $10^9 M_\odot$ of molecular hydrogen can be found in early-type galaxies as well.

Astronomers have also discovered that perhaps 50% of all elliptical galaxies contain appreciable amounts of dust, with typical masses in the range of 10^5 to $10^6 M_\odot$. One intriguing observation is that there exist dust lanes that are essentially randomly oriented with respect to their parent galaxy's optical-light axes. Frequently, the dust is even found to be counter-rotating relative to the rotation direction of other (stellar) components. Apparently, much of this gas was acquired after the initial formation of the galaxy. More will be said about the dynamics of dust rings in ellipticals in Section 26.1.

Metallicity Gradients and Color Gradients

As we saw in Fig. 25.12 during our discussion of spirals, the metallicity of elliptical galaxies is well correlated with luminosity; brighter galaxies have higher overall metal content.

Furthermore, as with spiral galaxies, metallicity gradients and color gradients also exist in ellipticals. Generally, the central regions of E's are redder and more metal-rich than are regions at larger radii. This trend can be expressed in terms of the gradient of the mass fraction of metals found in the distribution of stars; typically $d \log_{10} Z / d \log_{10} r \approx -0.13$. The same is also true for lenticular galaxies. In addition, the disks of S0's are bluer than their bulges, as is the case for galaxies of later Hubble type [for S0's, $(U - V)_{\text{disk}} - (U - V)_{\text{bulge}} \sim -0.1$ to -0.5]. Any successful theory of galaxy formation must incorporate the available observations concerning chemical enrichment.

The Faber–Jackson Relation

One relationship that dE's, dSph's, normal E's, and the bulges of spirals all have in common is the correlation between their central radial-velocity dispersion and M_B . This relation is shown in Fig. 25.33. To see why the relation arises, we begin with the result of the virial theorem for the central radial-velocity dispersion of a spherically symmetric mass distribution (Eq. 25.13). In that case we made the simple approximation that a mass M was uniformly distributed throughout a volume of radius R . Now, if we further assume that the mass-to-light ratio is essentially constant for all galaxies and that their average surface brightnesses are all equal (assumptions that were also made in “deriving” the Tully–Fisher relation), we arrive at the result

$$L \propto \sigma_0^4, \quad (25.40)$$

where σ_r has been replaced by σ_0 to signify that the central value for the radial-velocity dispersion is being assumed. This relationship was first identified by Sandra Faber and Robert Jackson and is now referred to as the **Faber–Jackson relation**. Expressing the luminosity in solar units, taking the logarithm of both sides of the equation, and writing

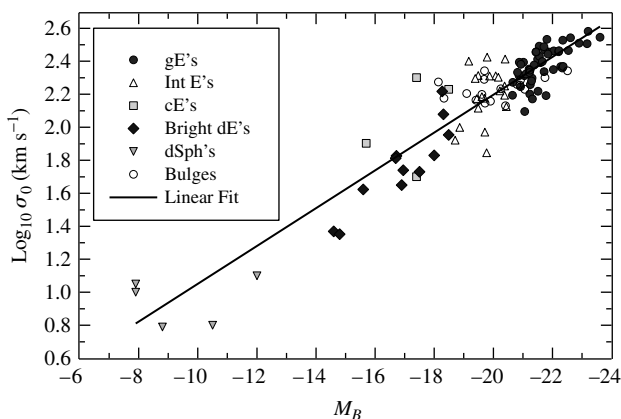


FIGURE 25.33 The Faber–Jackson relation represents a correlation between central velocity dispersion and luminosity. The relation is a consequence of the virial theorem. (Data from Bender et al., *Ap. J.*, 399, 462, 1992.)

$\log_{10}(L/L_{\odot})$ as a difference in M_B 's via Eq. (3.8), we find the linear relationship between $\log_{10} \sigma_0$ and M_B illustrated in Fig. 25.33, namely

$$\log_{10} \sigma_0 = -0.1 M_B + \text{constant}. \quad (25.41)$$

[Note the resemblance to the Tully–Fisher relations given by Eqs. (25.5–25.7) if those expressions are solved for $\log_{10} V_{\text{max}}$.]

The Fundamental Plane

As is readily apparent by inspecting Fig. 25.33, there is considerable scatter in the data. This is further reflected in the fact that the slope of the best-fit line through the data differs slightly from the one found in Eq. (25.41). Apparently Eq. (25.40) is not strictly true, which should not be surprising given the assumptions made in its development. Depending on the sample set used, $L \propto \sigma_0^{\alpha}$, with $3 < \alpha < 5$.

In an effort to find a tighter fit to the data, astronomers have introduced a second parameter into the expression, the effective radius. One representation of this fit is

$$L \propto \sigma_0^{2.65} r_e^{0.65}. \quad (25.42)$$

Here, galaxies are visualized as residing on a two-dimensional “surface” in the three-dimensional “space” represented by the coordinates L , σ_0 , and r_e . Known as the **fundamental plane**, Eq. (25.42) combines the contributions of a galaxy’s gravitational well (σ_0) with its radius and luminosity. Alternatively, the fundamental plane can be written in terms of the effective surface brightness of a galaxy at its effective radius:

$$r_e \propto \sigma_0^{1.24} I_e^{-0.82}. \quad (25.43)$$

The fundamental plane appears to represent the whole family of elliptical galaxies and seems to imply something about the formation of these systems.

The Effects of Rotation

Parameterizations such as the fundamental plane do a much better job of classifying ellipticals than Hubble’s original scheme based on ellipticity. But then, what is the source of a galaxy’s shape? Although the answer is not yet entirely clear, it is evident that most ellipticals are not purely oblate or prolate rotators with two axes, but are triaxial, meaning that there is no single preferred axis of rotation. We have already encountered this situation in the randomness of the dust lanes found in at least one-half of all ellipticals. Further evidence exists in the frequent observations of counter-rotating stellar cores in as many as 25% of larger ellipticals. It appears that in at least some of these cases, material in the form of gas, dust, globular clusters, or dwarf galaxies has been captured sometime since the galaxy’s formation. This situation will be discussed in more detail in Section 26.1.

When the rotational velocities of elliptical galaxies are measured, we find that the more luminous ones have mean rotational velocities that are much less than their velocity dispersions, implying that the shapes of these galaxies are not due to rotation. Instead, their

shapes are caused by the anisotropic velocity dispersion of stars in the galaxies.²¹ In one particularly extreme case, the luminous E4 galaxy, NGC 1600 ($M_B = -22.87$), has $V_{\text{rot}} = 1.9 \pm 2.3 \text{ km s}^{-1}$ and $V_{\text{rot}}/\sigma < 0.013$; it has no unique axis of rotation that is statistically significant.

If the ellipticity, ϵ (Eq. 25.1), of a galaxy were due to its being an ideal, oblate rotator with an isotropic stellar velocity distribution, then it can be shown that

$$\left(\frac{V_{\text{rot}}}{\sigma}\right)_{\text{isotropic}} \approx \left(\frac{\epsilon}{1-\epsilon}\right)^{1/2}.$$

For NGC 1600 this means that if its ellipticity of $\epsilon = 0.4$ were due to pure rotation, then V_{rot}/σ should be approximately 0.8. Defining the rotation parameter $(V/\sigma)^*$ to be

$$(V/\sigma)^* \equiv \frac{(V_{\text{rot}}/\sigma)_{\text{observed}}}{(V_{\text{rot}}/\sigma)_{\text{isotropic}}}$$

gives $(V/\sigma)^* < 0.016$ for NGC 1600. Although the distinction is somewhat arbitrary, a galaxy is considered to be primarily **rotationally supported** if $(V/\sigma)^* > 0.7$.

Bright E's and gE's have typical values of $\langle (V/\sigma)^* \rangle \approx 0.4$ and are **pressure-supported**, meaning that their shapes are due primarily to random stellar motions. In addition, the diffuse, faint dwarf E's have predominantly anisotropic velocity dispersions and are also pressure-supported. On the other hand, galaxies for which $-18 > M_B > -20.5$ mag have $\langle (V/\sigma)^* \rangle \approx 0.9$, implying that they are largely rotationally supported (this includes the cE's). Interestingly, the bulges of spiral galaxies tend to be rotationally supported as well.

Correlations with Boxiness or Diskiness

In 1988 Ralf Bender, Jean-Luc Nieto (1950–1992), and their collaborators proposed that many of the characteristics of elliptical galaxies can be understood in terms of the degree of **boxiness** or **diskiness** that their isophotal surfaces exhibit. As a means of quantifying the deviation from an elliptical shape, the shape of an isophotal contour (defined for some specific value of μ) is written in polar coordinates as a Fourier series of the form

$$a(\theta) = a_0 + a_2 \cos(2\theta) + a_4 \cos(4\theta) + \dots, \quad (25.44)$$

where a is the contour's radius and the angle θ is measured counterclockwise from the major axis of the ellipsoid. The first term in the expansion represents the shape of a perfect circle, the second term corresponds to the amount of ellipticity, and the third term is associated with the degree of boxiness. If $a_4 < 0$, the isophotal surface tends toward a “boxy” appearance, and if $a_4 > 0$, the surface tends toward being “disky” (see Fig. 25.34). Typically, $|a_4/a_0| \sim 0.01$, with deviations from perfect ellipses routinely measured to an accuracy of approximately 0.5%.

Figure 25.35 shows the correlations of various quantities with the isophotal Fourier parameter a_4 , measured in units of $100a_4/a_0$. The upper left-hand panel illustrates that disky galaxies tend to be rotationally supported [with larger values of $(V/\sigma)^*$], while boxy

²¹An anisotropic velocity distribution is one for which one or two directions of motion are preferred, rather than being completely random in all three dimensions, as in an isotropic distribution.

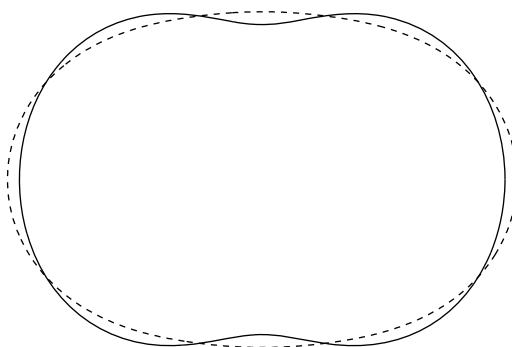


FIGURE 25.34 Two examples of galaxies with $\epsilon = 0.4$. $|a_4/a_0|$ has been set to the rather large value of 0.03 to illustrate the effect. The solid line represents a “boxy” galaxy ($a_4/a_0 < 0$), and the dashed line represents a “disky” galaxy ($a_4/a_0 > 0$).

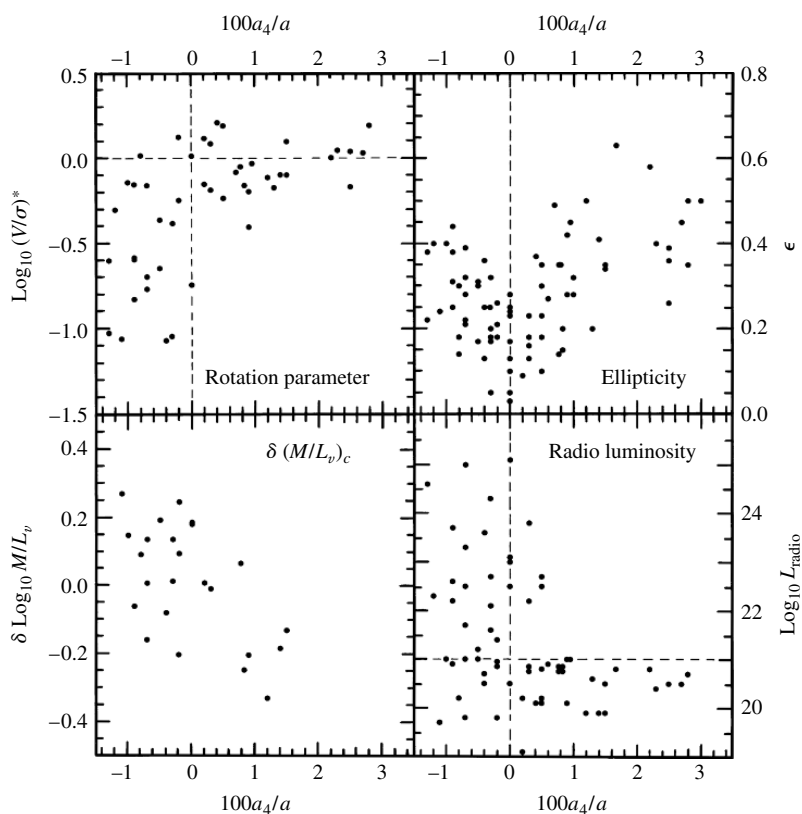


FIGURE 25.35 The correlations of various parameters with a_4 (isophotal shape). “Disky” ellipticals have $a_4 > 0$, while “boxy” ellipticals have $a_4 < 0$. (Figure adapted from Kormendy and Djorgovski, *Annu. Rev. Astron. Astrophys.*, 27, 235, 1989. Reproduced with permission from the *Annual Review of Astronomy and Astrophysics*, Volume 27, ©1989 by Annual Reviews Inc.)

galaxies are largely pressure-supported. The lower left-hand panel indicates the deviation of the core mass-to-light ratio (measured in the V band) from the average value for galaxies of the same luminosity; boxy galaxies tend to have higher-than-average ratios, while disk galaxies have lower-than-average values. The panel in the lower right-hand corner shows that boxy ellipticals tend to be much brighter at radio wavelengths than are disk galaxies. However, the spread in the radio luminosities of boxy galaxies is quite large, while the variation in radio luminosity for disk galaxies is much smaller. The same sort of correlation is found in other wavelength regions as well (such as X-ray, UV, and $H\alpha$). For disk E's, the sources of X-rays tend to be consistent with compact sources only, as is the case for S0's, while boxy E's show evidence of diffuse sources (i.e., hot gas). Finally, the upper right-hand panel indicates that ellipticity (and therefore Hubble type) tends to increase with the absolute value of a_4 ($|a_4|$). However, this suggests that two galaxies of identical Hubble type could have very different characteristics when other physical parameters are considered. For instance, for two E4 galaxies ($\epsilon = 0.4$), one could be rotationally supported ($a_4 > 0$) while the other is pressure-supported ($a_4 < 0$), and the former galaxy would probably have a core mass-to-light ratio lower than average while the latter galaxy would have a higher-than-average core mass-to-light ratio. Also, the former galaxy would have a low radio luminosity, and the latter might have a much higher radio luminosity.

Other parameters also seem to be associated with the boxiness or diskiness of a galaxy. Whereas rotationally supported disk galaxies have relatively large ratios of rotational angular momentum to mass (specific angular momentum), pressure-supported boxy galaxies have rather low ratios. Boxy galaxies are frequently observed to contain counter-rotating cores, while disk galaxies rarely possess counter-rotating cores. Finally, disk galaxies generally have rotational symmetries that are oblate, and boxy galaxies tend to be much more triaxial.

Although it is likely that the degree of boxiness among galaxies represents a continuum rather than two distinctly different classes of objects, apparently some 90% of all ellipticals are generally disk in nature. It has been suggested that disk galaxies may simply represent an extension of the sequence of S0's to progressively smaller and smaller disk-to-bulge ratios, and that some disk galaxies are actually misclassified S0's. However, boxy galaxies may well represent a tell-tale signature of some level of environmental evolution, such as tidal interactions or mergers (Section 26.1).

The Relative Numbers of Galaxies of Various Hubble Types

Before leaving the discussion of morphology, it is important to address the question of the relative numbers of galaxies of various Hubble types. This is usually represented by the **luminosity function**, $\phi(M) dM$, defined to be the number of galaxies in a particular sample that have absolute magnitudes between M and $M + dM$. Figure 25.36 shows luminosity functions for two samples; the top graph is for galaxies located in the vicinity of the Milky Way and the bottom graph is for a sample of galaxies in the Virgo cluster of galaxies (see Section 27.3). The total luminosity function in either environment is the sum of the individual luminosity functions of each Hubble type.

In an attempt to find a general analytic fit to galactic luminosity functions, Paul Schechter proposed the functional form

$$\phi(L) dL \sim L^\alpha e^{-L/L^*} dL, \quad (25.45)$$

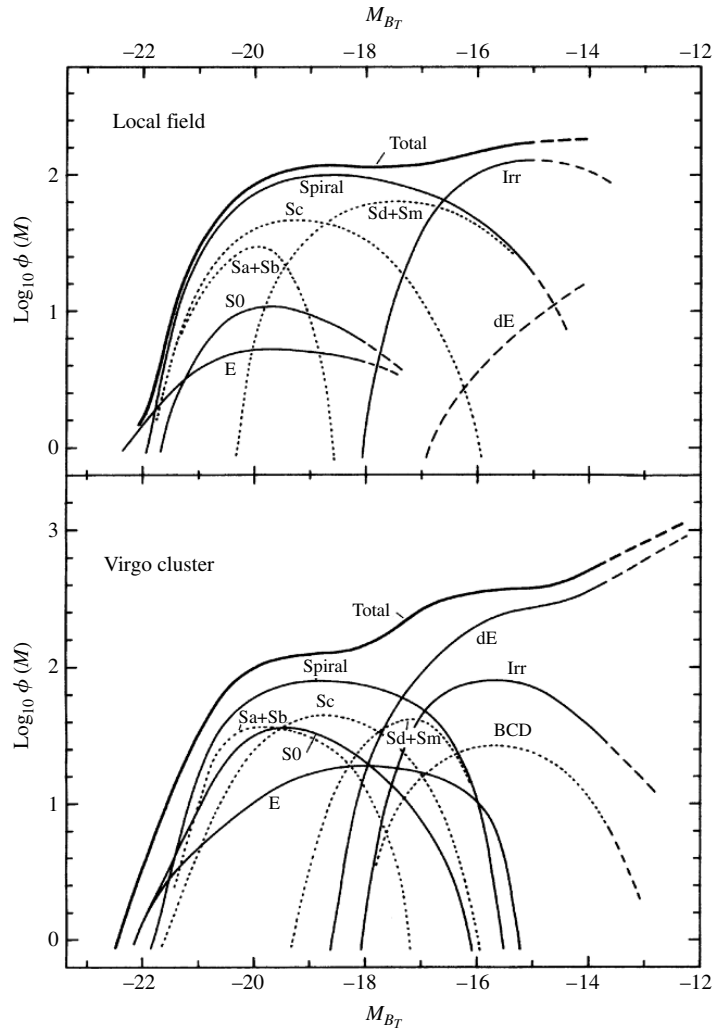


FIGURE 25.36 Luminosity functions for two samples of galaxies. The top panel is based on a sample of galaxies near the Milky Way Galaxy, and the bottom panel is for galaxies located in the Virgo cluster of galaxies. The zero point of the $\log_{10} \phi(M)$ axis is arbitrary. (Figure adapted from Binggeli, Sandage, and Tammann, *Annu. Rev. Astron. Astrophys.*, 26, 509, 1988. Reproduced with permission from the *Annual Review of Astronomy and Astrophysics*, Volume 26, ©1988 by Annual Reviews Inc.)

which can also be written in the equivalent form

$$\phi(M) dM \sim 10^{-0.4(\alpha+1)M} e^{-10^{0.4(M^*-M)}} dM. \quad (25.46)$$

In both forms, α and L^* (or M^*) are free parameters that are used to obtain the best possible fit to the available data. For the data in Fig. 25.36, $\alpha = -1.0$ and $M_B^* = -21$ for the local

field of galaxies near the Milky Way, while $\alpha = -1.24 \pm 0.02$ and $M_B^* = -21 \pm 0.7$ for the Virgo cluster. Although the values are similar, it is clear that there is not a universal luminosity function. Instead, each luminosity function depends on the environment of the particular sample of galaxies.

As can be seen for either environment depicted in Fig. 25.36, dE's and dwarf irregulars represent the largest fraction of all galaxies, even though S's and E's are the most prominent in terms of both cluster luminosity and mass. Careful inspection of the figures also shows another important detail: Although spirals represent the largest fraction of bright galaxies in each case, there is a somewhat higher proportion of ellipticals in the Virgo cluster. When the Virgo cluster is compared with the much larger and more densely populated Coma cluster, the relative numbers of spirals and ellipticals change dramatically. In the Virgo cluster we find 12% E's, 26% S0's, and 62% S+Ir, while in the Coma cluster, the proportions are 44% E's, 49% S0's, and only 7% S+Ir. Once again there is evidence that environment plays a role in galaxy formation and/or evolution, an issue we will explore in more detail in Chapter 26.

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PROBLEMS

- 25.1** Using Shapley's assumption that M101 has a diameter of 100 kpc, and adopting van Maanen's flawed observation of a measurable rotational proper motion, estimate the speed of a point at the edge of the galaxy and compare it to the characteristic rotation speed of the Milky Way.

- 25.2** (a) The absolute magnitude of M101, an Sc galaxy, is -21.51 in the B band. Using Eq. (25.11), estimate its isophotal radius (R_{25}) at $25\ B\text{-mag arcsec}^{-2}$.
 (b) Use the Tully–Fisher relation (Eq. 25.7) to estimate the rotational velocity of M101.
 (c) Estimate the angular rotation speed of a star at R_{25} , in units of arcsec yr^{-1} .
 (d) Could van Maanen have detected the rate of rotation of M101? How long would it take for the galaxy to rotate through $1''$?

- 25.3** (a) From the data given in Table 24.1, estimate M_B for our Galaxy. *Hint:* Be sure to include M_B for the Sun in your calculations, not M_{bol} ; see Appendix G.
 (b) Using the Tully–Fisher relation, calculate the maximum rotation speed of the Galactic disk and compare your answer with the observational value given in Section 24.3.

- 25.4** Neglecting the effects of extinction and the K -correction, show that the surface brightness of a galaxy is independent of its distance from the observer.

- 25.5** Prove that Eq. (25.2) follows directly from Eq. (24.13).

- 25.6** Prove that Eq. (25.3) follows directly from Eq. (24.10).

- 25.7** (a) Show that Eq. (24.13) can be written as

$$I(r) = I_e e^{-b[(r/r_e)^{1/4} - 1]},$$

where $b = 7.67$.

- (b) Integrating over the entire surface brightness profile, show that the luminosity of the galaxy can be expressed in terms of r_e and I_e as

$$L_{\text{tot}} = \int_0^\infty 2\pi r I(r) dr = 8! \frac{e^{7.67}}{(7.67)^8} \pi r_e^2 I_e \simeq 7.22 \pi r_e^2 I_e. \quad (25.47)$$

Hint: $\int_0^\infty e^{-x} x^7 dx = \Gamma(8) = 7!$.

- (c) Recalling that $I_e \equiv I(r_e)$, show that if the integration of Eq. (25.47) is carried over $0 \leq r \leq r_e$ rather than over $0 \leq r < \infty$, then the resulting luminosity is $\frac{1}{2} L_{\text{tot}}$, consistent with the definition of r_e .
- 25.8** NGC 2639 is an Sa galaxy with a measured maximum rotational velocity of $324\ \text{km s}^{-1}$ and an apparent magnitude of $B = 12.22\ \text{mag}$ (after making corrections for extinction).
 (a) Estimate its absolute magnitude in the B band from the Tully–Fisher relation.
 (b) Determine the distance to NGC 2639 using its distance modulus.
 (c) What is the galaxy's radius (R_{25}) at a surface brightness level of $25\ B\text{-mag arcsec}^{-2}$?
 (d) Find the mass of NGC 2639 that is interior to R_{25} .
 (e) What is the luminosity of the galaxy in the B band? (Refer to the hint in Problem 25.3.)
 (f) Calculate the mass-to-light ratio for NGC 2639 in the B band, interior to R_{25} .
- 25.9** Referring to the color indices ($\langle B - V \rangle$) given in Table 25.1 and Appendix G, estimate the *average* (or integrated) spectral classification of main-sequence stars in spiral galaxies of types Sa, Sb, and Sc.
- 25.10** Use the rotation curve data in Fig. 25.14 to estimate the mass within the central $1''$ of the center of M32. Compare your answer with the value obtained using the velocity dispersion data in Example 25.2.1 and with the estimated range quoted in that example.

- 25.11** (a) From the data shown in Fig. 25.37 for the stellar rotational velocities near the center of M31, estimate the amount of mass within $1''$ of the center of the galaxy. Compare your answer with the value quoted in the text.
- (b) Estimate the amount of mass within the central $1''$ based on the velocity dispersion data.
- (c) Comment on the source of the asymmetries evident in Fig. 25.37. Recall Fig. 25.13.

25.12 Beginning with the general expression for the position vector in rectangular coordinates

$$\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}},$$

show that the vector can be represented in cylindrical coordinates by Eq. (25.16).

25.13 Show that the acceleration vector is given by Eq. (25.18) in cylindrical coordinates. *Hint:* Note that the unit vectors $\hat{\mathbf{e}}_R$ and $\hat{\mathbf{e}}_\phi$ are position-dependent and therefore time-dependent. You may find their relationships with rectangular-coordinate unit vectors helpful.

25.14 Using the solar motion data given in Chapter 24, estimate the amplitude of the Sun's excursion in the radial direction relative to a perfectly circular orbit. Assume that the Sun is currently at the midpoint of its oscillation. Does your result represent a minimum or a maximum estimate of the actual deviation?

- 25.15** (a) From the information given in the text, derive Eq. (25.37) for the square of the solar epicycle frequency.
- (b) Show that Eq. (25.38) follows directly from Eq. (25.37).

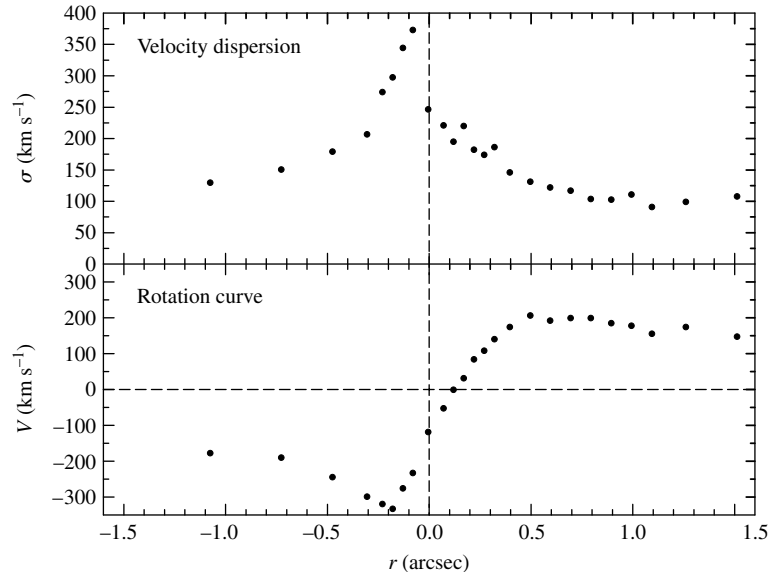


FIGURE 25.37 The stellar velocity dispersion and rotational velocities of stars near the center of M31, measured along the major axis of the bulge. Given the distance to Andromeda of 770 kpc, $1''$ corresponds to a linear distance from the center of 3.7 pc. (Data from Bender et al., *Ap. J.*, 631, 280, 2005.)

25.16 Determine the epicycle axis ratio, $\chi_{\max}/\rho_{\max}$, for a Keplerian orbit (a point mass orbiting a central, spherically symmetric mass). *Hint:* Begin with Eq. (25.37).

25.17 Show that if the surface brightness of an elliptical galaxy follows the $r^{1/4}$ law given by Eq. (24.13), then the average surface brightness over the area of a circular disk of radius r_e is given by

$$\langle I \rangle = 3.607 I_e.$$

Hint: Begin by rewriting the $r^{1/4}$ law in the form

$$I(r) = I_e e^{-\alpha[(r/r_e)^{1/4} - 1]}$$

and recall the definition of an integral average given in Problem 2.9. You may also find it helpful to write your integral in such a way that the limits of integration extend from zero to infinity. This can be done by considering the definition of r_e .

25.18 Show that the Holmberg radius of a galaxy obeying the $r^{1/4}$ law is related to the galaxy's effective radius, r_e , and the corresponding isophotal surface brightness, μ_e , by

$$r_H = r_e (4.18 - 0.12 \mu_e)^4.$$

25.19 (a) Use the result of Problem 25.17 to show that $\langle \mu \rangle = \mu_e - 1.393$ for an elliptical galaxy that follows the $r^{1/4}$ law.

(b) NGC 3091 has an effective radius of 10.07 kpc in the B band and an average surface brightness within the effective radius of 21.52 B -mag arcsec $^{-2}$. From this information, determine μ_e in the B band.

(c) What is the Holmberg radius of NGC 3091? You may find the result of Problem 25.18 useful.

25.20 According to the virial theorem, the central radial-velocity dispersion is related to the mass and size of the galaxy by $\sigma_r^2 \propto M/R$ (see Eq. 25.13). Use arguments similar to those for the Tully–Fisher relation to show that $L \propto \sigma_r^4$, which is the Faber–Jackson relation, Eq. (25.40).

25.21 (a) From the data given in Fig. 25.33, estimate the slope of the curve that represents the best-fit linear relationship.

(b) How does the slope in Fig. 25.33 compare with Eq. (25.41)? Why wouldn't you expect them to be exactly the same?

25.22 (a) It is estimated that M31 has approximately 350 globular clusters. If its absolute visual magnitude is -21.7 , estimate the specific frequency for its clusters.

(b) NGC 3311 is a cD galaxy with an estimated 17,000 globular clusters and an absolute visual magnitude of -22.4 . Estimate the specific frequency of clusters in this galaxy.

(c) Discuss the problem of globular cluster statistics in the suggestion that cD galaxies are due to mergers of already formed spiral galaxies.

25.23 (a) Find a general expression for a_2 , the coefficient of the first-order Fourier term in Eq. (25.44), written in terms of a_0 and ϵ (see Eq. 25.1). Assume that all higher-order terms are identically zero for this part of the problem.

(b) Make a polar-coordinate plot of a as a function of θ for an E4 galaxy with $a_0 = 30$ kpc and a_2 determined from the relationship found in part (a). Again assume that all higher-order terms are identically zero.

- (c) Make a polar-coordinate plot for the same E4 galaxy, but with $a_4 = 0.1a_0$.
 - (d) Make a polar-coordinate plot for the same E4 galaxy, but with $a_4 = -0.1a_0$.
 - (e) Comment on the general appearance of your last two plots. Which one looks more like a lenticular galaxy?
- 25.24** Plot $\log_{10} \phi(M)$, the logarithm of the Schechter luminosity function, for both the local field of galaxies near the Milky Way and the Virgo cluster over the range $-23 < M_B < -12$ (see Eq. 25.46). Use the values of α and M^* given in the text. To compare your results with those given in Fig. 25.36, shift your data so that $\log_{10} \phi(-23) = 0$ for both groups of galaxies.

CHAPTER

26

Galactic Evolution

26.1 *Interactions of Galaxies*

26.2 *The Formation of Galaxies*

26.1 ■ INTERACTIONS OF GALAXIES

With the exquisite vision of the heavens provided by modern ground-based and space-based observatories, it has become increasingly apparent that galaxies are not “island universes”; they do not evolve in isolation.

Evidence of Interactions

Nearly all galaxies belong to clusters, and the galaxies take up a larger fraction of the cluster’s volume than do the stars in a stellar cluster. We also know that the spacing between galaxies is typically only 100 times larger than the size of the galaxies themselves (see Section 27.3). Densely populated clusters, such as the Coma cluster (Figs. 26.1 and 26.2), have a higher proportion of early-type galaxies (ellipticals) in their centers than they do in their outer, less dense regions. As was already mentioned in Section 25.4, the central regions of these rich, regularly shaped clusters also have a higher proportion of early-type galaxies than the centers of less populated, amorphous-shaped irregular clusters, such as the Hercules cluster (Fig. 26.3). These observations seem to correlate with the increased probability of interactions and/or mergers between galaxies in regions of higher galaxy number density. Interactions tend to increase the velocity dispersions of stars in the galaxies involved, possibly destroying disk structures in late-type galaxies and causing the galaxies to relax to early-type $r^{1/4}$ profiles (Eq. 25.2).

A VLA radio survey of the H I layer of galactic disks found that at least 50% of all disk galaxies display warped disks. Also, more than half of all elliptical galaxies harbor discrete shells of stars.¹ Some disk warping may be due to tidal interactions with smaller satellite galaxies, and, as we shall see, shells in ellipticals are signatures of mergers.

Observations also suggest that hot, X-ray-emitting gas occupies much of the space between the galaxies in rich clusters and has a mass equal to or exceeding the mass of all of the cluster’s stars (Section 27.3). It seems that the gravitational influence of interacting galaxies is largely responsible for removing the gas from the individual galaxies that make up the cluster, while still leaving the gas trapped in the cluster’s overall gravitational well.

¹ See Binney (1992) and Barnes and Hernquist (1992) for further details on these statistics.



FIGURE 26.1 The center of the Coma cluster. The width of this view is about 18 arcmin. (Courtesy of National Optical Astronomy Observatories.)

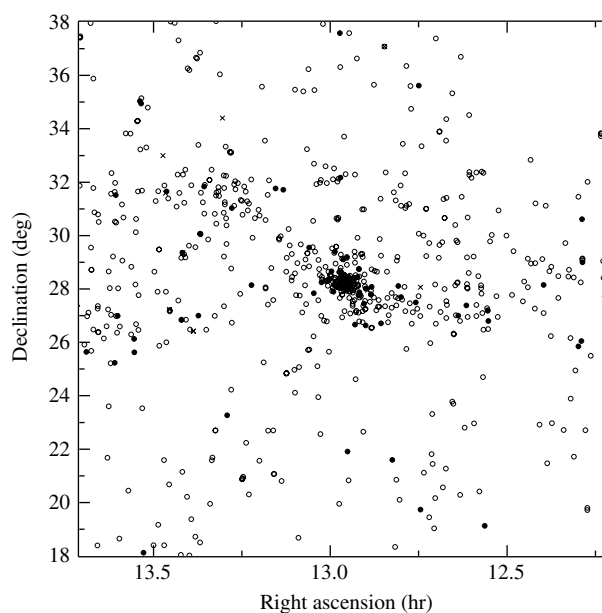


FIGURE 26.2 The Coma cluster of galaxies, showing the ellipticals (filled circles) and spirals (open circles). Note that the scale is much larger than the width of the image in Fig. 26.1.



FIGURE 26.3 The Hercules cluster of galaxies. (Courtesy of National Optical Astronomy Observatories.)

Although only a small fraction of the galaxies' gas is removed in a direct collision, mergers may initiate a burst of star formation that produces stellar mass loss and supernovae, leading to a galactic superwind capable of liberating a large amount of gas.

This evidence and that cited in Chapter 25 suggest that interactions between galaxies play an important role in their evolution. In the hierarchical “bottom-up” scenario of galaxy formation to be described in the next section, large galaxies are thought to be formed by mergers and the gravitational capture of smaller entities. In this view, the interactions seen today between galaxies are simply a natural extension of their formative years. Figure 26.4 shows one of many examples of galactic interactions. In this section, we will examine such obvious examples, as well as more subtle phenomena.

Dynamical Friction

What happens when galaxies collide? Given that stars are generally spread very far apart in galaxies, the chance of even a single stellar collision is quite small (see Problem 26.1). Instead, interactions between stars will be gravitational in nature. To see this, imagine that an object (a globular cluster or small galaxy) of mass M is moving through an infinite collection of stars, gas clouds, and dark matter with a constant mass density, ρ . We will assume that the mass of each object in the background “sea” of material is much less than M , so M continues moving in a straight line instead of being deflected. In the absence of collisions, it might be thought that M would move unimpeded. However, as M moves forward, the other objects are gravitationally pulled toward its path, with the closest ones feeling the largest force. As shown in Fig. 26.5, this produces a region of enhanced density along the path, with a high-density “wake” trailing M . The result, known as **dynamical friction**, is a net gravitational force on M that opposes its motion. Kinetic energy is transferred from M to the surrounding material as M 's speed is reduced.



FIGURE 26.4 The plumes on the opposite sides of NGC 520 are evidence of a tidal interaction, possibly ending in the merger of the two colliding disk galaxies. Note the diagonally oriented dust lane. (Courtesy of Gemini Observatory/AURA.)

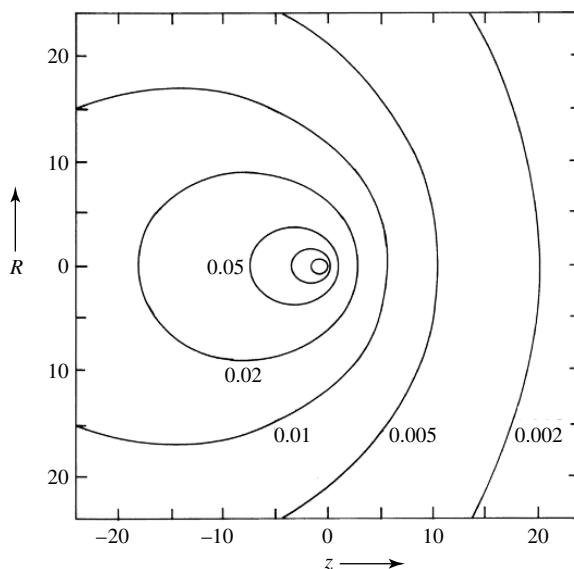


FIGURE 26.5 The fractional enhancement in the density of stars caused by the motion of a mass M in the positive z direction. (Figure adapted from Mulder, *Astron. Astrophys.*, 117, 9, 1983.)

The derivation of an expression for dynamical friction is beyond the scope of this text, but it is easy to see that the physical quantities involved are the mass, M , and speed, v_M , of the massive object, and the mass density of the surrounding material, ρ . As a result, the expression for the force must contain GM , v_M , and ρ only.² It is left as a problem to show that there is only one combination of these variables that has the units of force, so the expression for the force of dynamical friction, f_d , must look like

$$f_d \simeq C \frac{G^2 M^2 \rho}{v_M^2}, \quad (26.1)$$

where C is dimensionless. Here, C is not a constant, but a function that depends on how v_M compares with the velocity dispersion, σ , of the surrounding medium. For $v_M \sim 3\sigma$, typical values of C are 23 for the LMC, 76 for globular clusters, and 160 for ellipticals.³

Through a careful examination of Eq. (26.1), it is possible to see why the various terms enter as they do. Clearly the dynamical friction must be proportional to the mass density of stars. Assuming that the relative numbers of objects of various masses do not change, doubling ρ means doubling the total number of objects, which would in turn double the gravitational force on M . The mass M itself is squared; one power comes from its role in producing the high-density wake that trails behind it, and the other from the gravitational force on M produced by the enhanced density. Finally, consider the velocity-squared term in the denominator. If M moves twice as fast, it will spend only half as much time near a given object, and so the impulse $\int \mathbf{F} dt = \Delta \mathbf{p}$ given to that object is only half as great. Consequently, the density enhancement develops only half as rapidly, and M will be twice as far away by the time the enhancement arises. Thus the v_M^2 in the denominator comes from the inverse-square law of gravity. This last point means that slow encounters are much more effective at decreasing the speed of an intruding mass.

To get an estimate of the timescale associated with the effects of dynamical friction acting on a galaxy's globular clusters, recall that flat rotation curves imply that the density of the dark matter halo may be approximated most simply by Eq. (24.50),

$$\rho(r) = \frac{v_M^2}{4\pi G r^2}.$$

Inserting this expression into Eq. (26.1), we find that the dynamical friction acting on a cluster is

$$f_d = C \frac{G^2 M^2 \rho(r)}{v_M^2} = C \frac{G M^2}{4\pi r^2}.$$

If the cluster's orbit is circular and of radius r , its orbital angular momentum is just $L = M v_M r$. Since dynamical friction acts tangentially to the orbit and opposes the cluster's

²Because the only time that M arises is in connection with Newton's law of gravity, the combination GM will be inseparable in any derivation.

³You may refer to Binney and Tremaine (1987) for a detailed discussion of dynamical friction.

motion, a torque of magnitude $\tau = r f_d$ is exerted on the cluster. The torque in turn reduces the cluster's angular momentum according to

$$\frac{dL}{dt} = \tau.$$

Recalling that a flat rotation curve implies that the orbital speed v_M is essentially constant, differentiating the angular momentum and substituting for the torque give

$$M v_M \frac{dr}{dt} = -r C \frac{G M^2}{4\pi r^2}.$$

Integrating the last equation leads to an expression describing the time required for the globular cluster to spiral into the center of the host galaxy from an initial radius r_i , or

$$\int_{r_i}^0 r dr = -\frac{C G M}{4\pi v_M} \int_0^{t_c} dt.$$

Solving for the cluster lifetime, t_c , yields

$$t_c = \frac{2\pi v_M r_i^2}{C G M}. \quad (26.2)$$

Equation (26.2) can be inverted to find the most distant cluster that could have been captured within the estimated age of the galaxy:

$$r_{\max} = \sqrt{\frac{t_{\max} C G M}{2\pi v_M}} \quad (26.3)$$

Example 26.1.1. Consider a globular cluster that orbits the Andromeda galaxy (M31). Assume that the cluster's mass is $M = 5 \times 10^6 M_\odot$ and its velocity is $v_M = 250 \text{ km s}^{-1}$, typical of the rotation curve in the outer part of the galaxy. If the age of M31 is approximately 13 Gyr, then Eq. (26.3) implies that $r_{\max} = 3.7 \text{ kpc}$. This means that globular clusters of $5 \times 10^6 M_\odot$ that were originally within approximately 4 kpc of the center of Andromeda would have spiraled into its nucleus by now.

According to Eq. (26.3), $r_{\max} \propto M^{1/2}$, implying that clusters with masses greater than $5 \times 10^6 M_\odot$ could have been gathered from greater distances. This may help to explain why there are no very massive globular clusters remaining around M31 today.

Not only are globular clusters affected by dynamical friction, but satellite galaxies are as well. Recall that on page 891 we learned that a stream of material has been tidally stripped from the Magellanic Clouds. In fact, it appears that dynamical friction will ultimately cause the Magellanic Clouds to merge with the Milky Way some 14 billion years in the future. This is a fate that has already befallen the Sagittarius dwarf spheroidal galaxy, the remnant of the dwarf galaxy in Canis Major, and possibly a progenitor dwarf galaxy of ω Centauri. In fact, any giant galaxy will probably devour numerous satellite galaxies during its lifetime.

The process of satellite accretion has a variety of possible consequences. For instance, the gravitational torques involved in the merger of a satellite galaxy in a retrograde orbit may produce the counter-rotating cores that are observed in some elliptical galaxies (recall Section 25.4). In a disk galaxy, mergers may excite global modes of oscillation (similar to those in pulsating stars; see Chapter 14). Mergers are also capable of producing the sort of featureless disks that are characteristic of Ir (amorphous) galaxies. The importance of mergers in the evolution of galaxies will be explored in Section 26.2.

Rapid Encounters

We now turn to another type of encounter, one that occurs so rapidly between two galaxies that their stars do not have time to respond. Even in the special case where the two systems pass through one another, there is no significant dynamical friction because there is no appreciable density enhancement. In this *impulse approximation*, the stars barely have time to alter their positions. As a result, the internal potential energy, U , of each galaxy is unchanged by the collision. However, the gravitational work that each galaxy performs on the other has increased the internal kinetic energies of both galaxies in a random way. This internal kinetic energy comes at the expense of the overall kinetic energies of the galaxies' motions with respect to one another. The amount of internal energy gained by the galaxies depends on the nearness of the approach of the galaxies involved, with the energy declining rapidly for more distant encounters. (Roughly, the energy gained is inversely proportional to the fourth power of the distance of closest approach of the galaxies.)

Suppose that one of the galaxies gains some internal kinetic energy during the interaction. To determine how it will respond, assume that the galaxy was initially in equilibrium, meaning that it satisfied the virial theorem prior to the encounter (recall Eqs. 2.46 and 2.47), and its initial kinetic, potential, and total energies were related by $2K_i = -U_i = -2E_i$. Imagine that during the encounter, the galaxy's internal kinetic energy increased from K_i to $K_i + \Delta K$. Because its potential energy has remained essentially constant, the galaxy's total energy has increased to $E_f = E_i + \Delta K$. As a result, the galaxy has been thrown out of virial equilibrium. When equilibrium is finally reestablished (after a timescale of a few orbital periods), the final kinetic energy must, according to the virial theorem, be

$$K_f = -E_f = -(E_i + \Delta K) = K_i - \Delta K.$$

Thus, as equilibrium is regained, the internal kinetic energy of the galaxy actually *decreases* by $2\Delta K$ from the value it had just after the collision.

How does the galaxy accomplish this reduction? One way to regain equilibrium is to convert the excess kinetic energy into an increased (less negative) gravitational potential energy. The galaxy expands slightly as the separation between its masses increases. Another way of reducing the kinetic energy is for the most energetic components to carry it out away from the galaxy in the form of a stream of stars and gas. This evaporation cools the galaxy and moves it toward a new equilibrium. In fact, both of these processes may occur, and which one dominates is determined by the specific circumstances of the collision. For instance, a high-speed, nearly head-on collision may produce a **ring galaxy**, such as the Cartwheel shown in Fig. 26.6. A numerical computer simulation and interpretation of this interaction are left as an exercise (Problem 26.20).

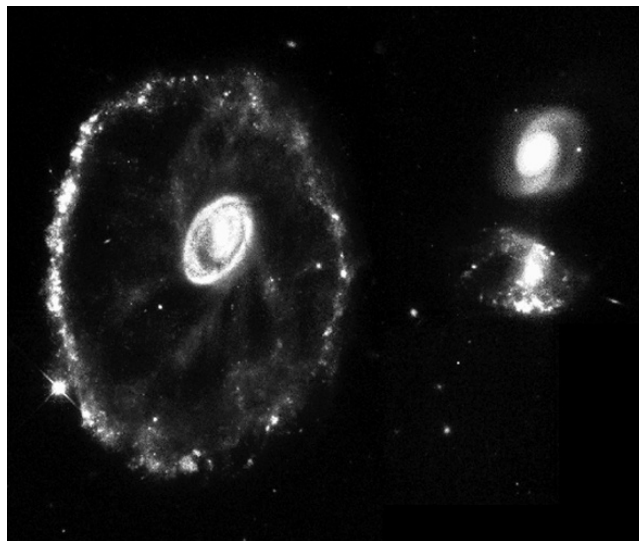


FIGURE 26.6 An HST image of the ring galaxy known as the Cartwheel (A0035–335). The diameter of the ring is about 46 kpc. As the ring expands at 89 km s^{-1} , it triggers bursts of star formation. It is not clear which of the two galaxies at right may have been the intruder. [Courtesy of Kirk Borne (STScI), and NASA.]

If two galaxies are gravitationally bound to one another, they will eventually merge, given enough time. Because of their extended mass distributions, galaxies do not follow the same trajectories they would if they were point masses; some orbital energy is diverted into increasing the galaxies' internal energy, and the orbit shrinks a bit. Tidal forces may also dissipate the orbital energy by pulling stars and gas out of one or both galaxies, a process known as **tidal stripping**. This is probably the cause of the Magellanic Stream.

An instructive way of thinking about this loss of galactic material involves using the idea of gravitational equipotential surfaces, discussed in Section 18.1 for a close binary system. Figure 18.3 shows the equipotential lines for a rotating system of two masses. Although the figure is for point masses in a circular orbit, it contains valuable insights for the present case of extended galaxies in noncircular orbits. In particular, it is useful to talk about the *tidal radii*, ℓ_1 and ℓ_2 , of the galaxies. They are the distances from the center of each galaxy to the inner Lagrangian point, L_1 , given by (approximately, in this noncircular case) Eqs. (18.9) and (18.10). As the two galaxies move about one another, the shape of the equipotential surfaces and the values of the tidal radii are constantly changing. If stars or clouds of gas extend beyond a galaxy's tidal radius, they have a tendency to escape from the gravitational potential well. (This is analogous to a star overflowing its Roche lobe, as discussed in Section 18.1.) As a result, the density of galactic material undergoes a sharp decline beyond the tidal radius.

Polar-ring galaxies and **dust-lane ellipticals** are normal galaxies that are orbited by rings of gas, dust, and stars that were stripped from other galaxies as they passed by or merged. Polar rings typically contain $10^9 M_\odot$ or more of gas and are found only around elliptical or S0 galaxies. It is estimated that some 5% of all S0 galaxies have or have had a

polar ring. Figure 26.7 shows the polar-ring galaxy NGC 4650A, an S0 galaxy with a ring that is nearly perpendicular to its long axis (inclined just 7°).

Because the rings respond to the gravitational field of the galaxies, astronomers use them as probes of the three-dimensional distribution of matter, both luminous and dark, in their host galaxies. The results of studying several polar-ring galaxies show that they are surrounded by spherical dark halos. An example of a well-known dust-lane elliptical is Centaurus A (NGC 5128), shown in Fig. 26.8. Cen A is also a powerful radio source and will be discussed in that context in Section 28.1.

An extreme case of a gaseous ring (no stars) is the one that encircles the central two galaxies of the M96 group in the constellation Leo, as shown in Fig. 26.9. The gas is in an

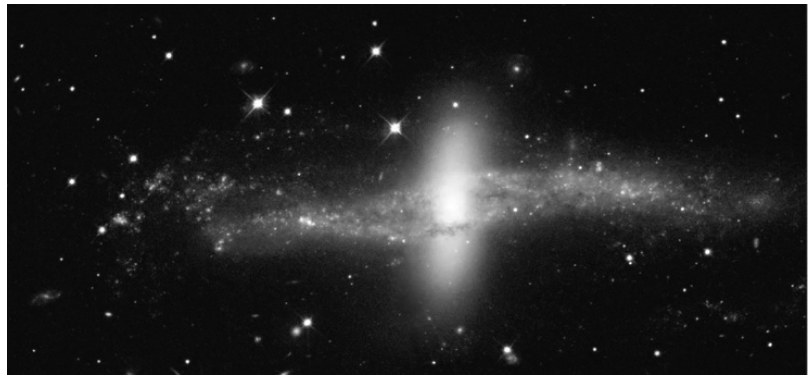


FIGURE 26.7 The polar-ring galaxy NGC 4650A. [Courtesy of J. Gallagher (University of Wisconsin–Madison) and the Hubble Heritage Team (AURA/STScI/NASA).]



FIGURE 26.8 Centaurus A (NGC 5128) is a dust-lane elliptical galaxy. (Courtesy of NOAO Cerro Tololo Interamerican Observatory.)

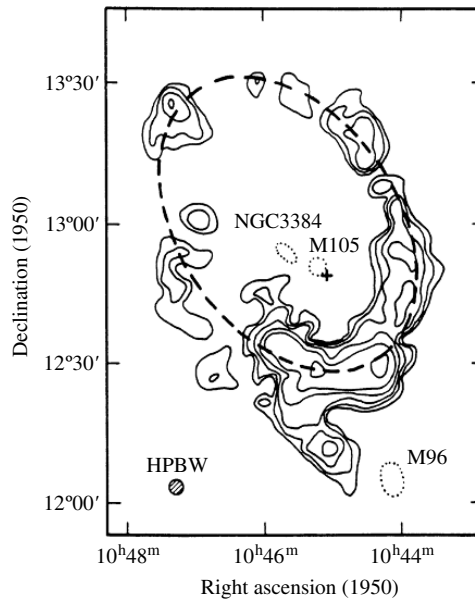


FIGURE 26.9 21-cm radio contours showing the ring of neutral hydrogen around the galaxies M105 (a giant elliptical) and NGC 3384 (an S0) in the M96 group. Note that the ring is more than 1° wide on the sky. (Figure adapted from Schneider, *Warped Disks and Inclined Rings around Galaxies*, Cambridge University Press, Cambridge, 1991.)

elliptical orbit around the two galaxies; the eccentricity of the orbit is $e = 0.402$, and the cross (“+”) in the figure marks the location of the orbit’s focus. Recall from Section 2.3 that an elliptical orbit is the result of an inverse-square force law, which means that the gas ring in the M96 group must lie *beyond* any dark-matter halos surrounding the two galaxies. The semimajor axis of the orbit is $a = 101$ kpc, so the maximum extent of any dark halo must be given by Eq. (2.5),

$$r_p = a(1 - e) = 60 \text{ kpc},$$

the perigalacticon distance. Note, however, that this calculation must be viewed cautiously because the ring is merely a locus of gas and may not in fact coincide with the orbit followed by a single particle.

Modeling Interactions with N -Body Simulations

The merger of two galaxies is a complicated affair. If the relative speed of the galaxies is substantially greater than the velocity dispersions of their stars, the collision will not result in a merger. Conversely, if the relative speed of the collision is less than the velocity dispersion of the stars in one of the galaxies, a merger is inevitable. The situation is particularly complicated when the relative velocity of the galaxies is comparable to the velocity dispersion of stars in one of the galaxies. This situation is best studied today by numerical experiments involving N -body simulations, like the ones that were used to investigate the

bar instability in a disk of stars (Section 25.3). In these computations, Newton's second law is used to follow the motions of the stars through a sequence of time intervals.⁴

One of the first such efforts was made in 1972 by Alar and Juri Toomre. Each galaxy consisted of a massive nucleus surrounded by concentric rings of disk stars. To simplify the calculations, the stars in the disk were allowed to feel the gravitational pull of the galactic nuclei, but not of each other. With their program, the Toomre brothers succeeded in demonstrating how interactions could explain the appearance of galaxies such as M51 (the Whirlpool), which has a bridge of stars and gas that appears to extend to its neighbor galaxy NGC 5195. Figure 25.17 shows this system, and Fig. 26.10 suggests how it could have been produced. Also note the development of the grand-design spiral arms that are so prominent in M51. In addition, the Toomres were successful in reproducing the appearance of **tidal-tail galaxies**, such as the pair NGC 4038 and NGC 4039 (the “Antennae”) shown in Fig. 26.11. It is remarkable that the rapid advances in computing power since the early 1970s now allow a program similar to the Toomres' to be run on a personal computer. Such a program, adapted from Schroeder and Comins (1988), is described in Appendix M.

In general, only close, slow encounters produce bridges and tails. The effect is most pronounced when the orbital angular speed of one of the galaxies matches the angular speed of some of the stars in the other galaxy's disk. The resulting orbital resonance, which acts on both the near and far sides of the disk, allows the tidal forces to be especially effective. Two bulges tend to develop on opposite sides of one (or both) of the galaxies, similar to Earth's tidal bulges. Tidal stripping then pulls out streams of stars and gas as the two galaxies pirouette around one another. When conditions are right, the stars and gas torn from the near side will form an apparent bridge, while, because of angular momentum conservation, the material stripped from the far side moves off to form a curving tail.

Modern computer codes include the effects of dark matter and the gravitational interactions (*self-gravity*) of individual stars and gas clouds. Figures 26.12 and 26.13 show the results of one such calculation that has reproduced the appearance of the tidal-tail galaxies NGC 4038 and NGC 4039. These codes have also been used to simulate the merger of a disk galaxy (with a central bulge and halo) and a satellite galaxy with 10% of the mass of the primary. The presence of a dark-matter halo significantly decreases the timescale of a merger because it allows the galaxies to interact over much larger distances. The satellite is devoured and absorbed in only two revolutions of the disk galaxy.

It is worth recalling that roughly one-half of the stars in the Milky Way's outer stellar halo move in retrograde orbits, which is probably a consequence of tidal stripping and/or mergers of other dwarf galaxies. Recall also (Section 24.2) that there are two different spatial distributions of globular clusters around the Milky Way. The inner clusters are old and vary little in age, while the outer population has a wider age spread. The inner clusters also tend to have a preferential direction for their orbits around the Galaxy, while the outer clusters have randomly oriented orbits. Again, it appears likely that these outer clusters formed elsewhere and were captured later by the Milky Way. In fact, some researchers

⁴In 1941, Holmberg used an *analog computer* to investigate the tidal interactions of galaxies. He made “galaxies” out of light bulbs and used the inverse-square law of light to mimic the same behavior for gravity. With the aid of a light meter, he was able to analyze the gravitational influence of one model galaxy on another.

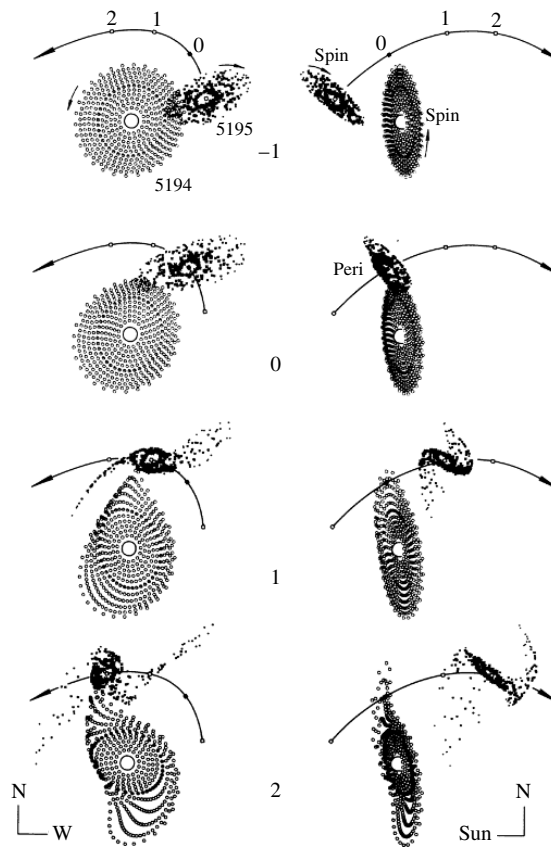


FIGURE 26.10 The Toomres' computer simulation of the interaction between M51 (NGC 5194) and NGC 5195. Note from the side view that the “bridge” does not actually connect the two galaxies. You may also note the resulting warped disk. [Figure adapted from Toomre, *The Large Scale Structure of the Universe*, Longair and Einasto (eds.), Reidel, Dordrecht, 1978.]

have suggested that the young globular clusters Palomar 12 and Ruprecht 106 were tidally stripped from the Large Magellanic Cloud.

Finally, if a satellite galaxy is moving at an angle with respect to the plane of the disk, the merger can result in a warped disk that persists for 3 to 5 billion years. More than 50% of galactic disks (including the Milky Way's) display a warp in their gas distributions (although some stars may participate in the warp as well), and most of these galaxies do not appear to have nearby companions. The absence of a companion might argue in favor of a merger, under the assumption that the companion was enveloped by the primary galaxy. However, most disks show no explicit evidence that a merger has occurred. It has also been shown that warped disks can arise simply as a consequence of the triaxial nature of the surrounding dark halos. The importance of the role of mergers or close encounters in producing warped disks has yet to be determined.



FIGURE 26.11 The Antennae galaxies, NGC 4038 and NGC 4039, and their tidal tails. [Courtesy of Brad Whitmore (STScI) and NASA.]

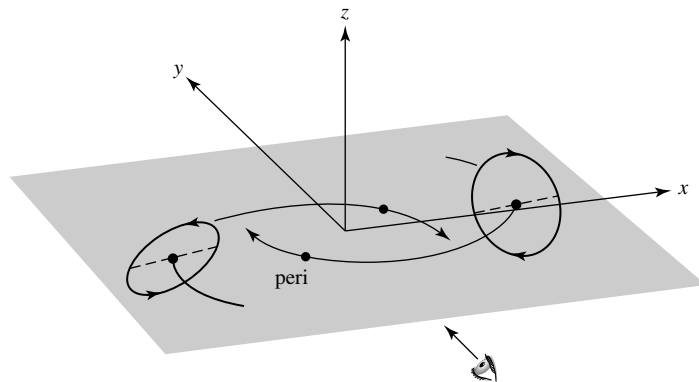


FIGURE 26.12 The orbital geometry used to calculate the formation of the tidal-tail galaxies shown in Fig. 26.11. The disks are shown in their initial positions. Each disk is initially inclined by 60° to the orbital plane, and a dashed line indicates the intersection of each disk with the orbital plane. The positions of closest approach are identified ("peri"). The viewing direction for Fig. 26.13 is indicated as along the positive y -axis. (Figure adapted from Barnes, *Ap. J.*, 331, 699, 1988.)

Starburst Galaxies

In 1972 Richard Larson and Beatrice Tinsley found that strongly interacting galaxies tend to be bluer than isolated ones of the same type. They attributed the excess blue light to hot newborn stars and argued that tidal interactions have induced vigorous bursts of star formation in these galaxies. The increased luminosity is difficult to detect because, like all star formation, it is shrouded in thick clouds of gas and dust (recall Section 12.2). The light emitted by the young stars at visible and ultraviolet wavelengths is absorbed and then reradiated in the infrared. In 1983 the Infrared Astronomy Satellite (IRAS) found that these

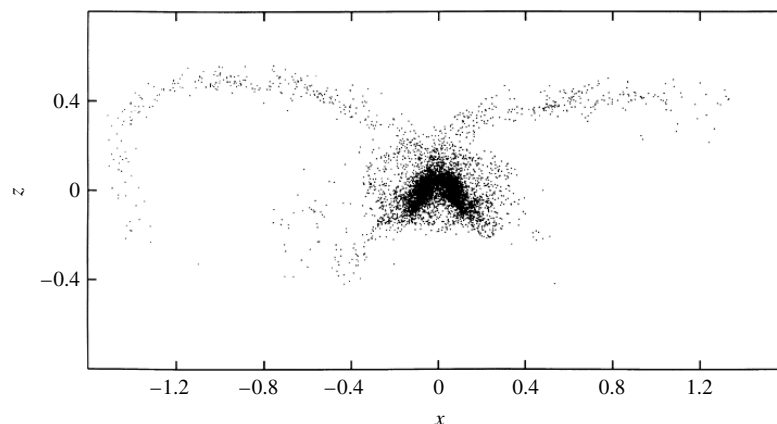


FIGURE 26.13 The result of Joshua Barnes's simulation of the formation of the tidal-tail galaxies NGC 4038 and NGC 4039. A distance of 1 on the x - and z -axes corresponds to 40 kpc. (Figure adapted from Barnes, *Ap. J.*, 331, 699, 1988.)

starburst galaxies are extremely bright at infrared wavelengths, pouring out up to 98% of their energy in this portion of the spectrum. (In comparison, the Milky Way emits 30% of its luminosity in the infrared, and M31 only a few percent.) Most of the star formation is confined to within about one kiloparsec or so of the galaxy's center, where radio telescopes have detected vast clouds containing some 10^9 to $10^{12} M_{\odot}$ of hydrogen gas that serves as fuel. Between 10 and 300 $M_{\odot}$ of gas is converted into stars each year in a starburst galaxy, while only two or three stars are formed annually in the Milky Way. The clouds typically contain sufficient hydrogen to support this rate of star formation for about 10^8 to 10^9 years, although a given burst may last for only 20 million years or so.

The observational evidence also shows that starbursts do not occur exclusively in the nuclei of interacting galaxies. Although this is often the case, many disk-wide starbursts are also known. In this case, the problem is how star formation can be triggered nearly simultaneously over such a wide area.

Since hydrogen clouds were initially distributed throughout the galactic disk, one puzzle facing astronomers is how a violent interaction with another galaxy could have removed more than 90% of the clouds' angular momentum, allowing the gas to become concentrated in the galactic center. An answer may be provided by numerical simulations of colliding galaxies. These studies indicate that the gas and stars react differently to the impact of an intruding galaxy, with the gas tending to move out in front of the stars as they orbit the galactic center. The stars' gravity then pulls back on the gas, and the resulting torque on the gas reduces its angular momentum, causing it to plunge toward the galactic center. As the two galaxies begin to merge, more angular momentum is lost. Shock fronts then compress the gas, and a burst of star formation begins.

Figure 26.14 shows the starburst galaxy M82/NGC 3034, about 3.2 Mpc away [see also Fig. 25.7(c)]. From its unusual appearance, astronomers once thought that this galaxy was exploding. The current interpretation is that between 10^7 and 10^8 years ago, the inner 400 pc of M82 began a tremendous episode of star formation that is still continuing. The center of M82 contains a wealth of OB stars and supernova remnants that have expelled more



FIGURE 26.14 The starburst galaxy M82 (NGC 3034). This composite image was obtained by the Subaru Telescope's FOCAS instrument in B , V , and $H\alpha$ wavelengths. [Courtesy of National Astronomical Observatory of Japan (NAOJ).]

than $10^7 M_{\odot}$ of gas from the plane of the galaxy. All of this violent activity may have been triggered by a tidal interaction with M81, a spiral galaxy that is now about 36 kpc away from M82 (as projected on the plane of the sky) and linked to it by a gaseous bridge of neutral hydrogen. In fact, the center of M81 is known to be a significant source of X-rays, with an X-ray luminosity of $L_X = 1.6 \times 10^{33}$ W. This has been interpreted as implying that M81 has a central black hole that is being “fed” by infalling gas at the rate of 10^{-5} to $10^{-4} M_{\odot} \text{ yr}^{-1}$. This gas infall may also be a consequence of the tidal interaction with M82.

Mergers in Elliptical and cD Galaxies

The importance of interactions in galactic evolution is most obvious for elliptical galaxies. The case is especially compelling for the cD galaxies typically found at the centers of rich, regular clusters. More than half of the known cD's have multiple nuclei that move differently than the galaxy as a whole. Although the typical orbital speed of a star in a cD galaxy is $\sim 300 \text{ km s}^{-1}$, the multiple nuclei move with relative velocities of $\sim 1000 \text{ km s}^{-1}$. All of this is taken as evidence that cD galaxies are the products of galactic mergers that supplied the supernumerary nuclei and the expansive halo of stars. Their frequent occurrence at the bottom of a cluster's gravitational potential well makes collisions and other close encounters all the more probable. Figure 26.15 is a remarkable photograph of three smaller galaxies passing through a cD at the center of the cluster Abell 2199. (In this case their velocities are so large that these galaxies will not be captured during this passage through the cD.) Recall from Fig. 25.16 and Table 25.3 that cD's also have unusually high specific frequencies of globular clusters, suggesting that a large number of clusters have been captured throughout the evolution of a typical cD galaxy.

Normal elliptical galaxies show evidence of interactions and mergers as well, particularly the boxy E's. As was mentioned in Section 25.4, about 25% of all ellipticals have very different velocity fields in their cores when compared with their outer regions. Observations

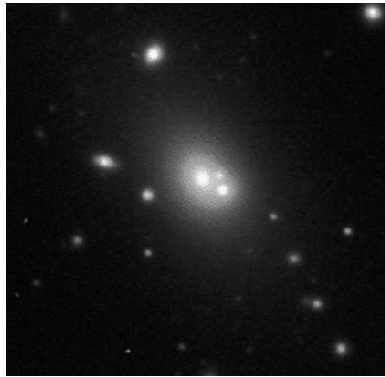


FIGURE 26.15 The passage of three small galaxies through a giant elliptical in the cluster Abell 2199. (Courtesy of Whipple Observatory and Harvard CfA.)

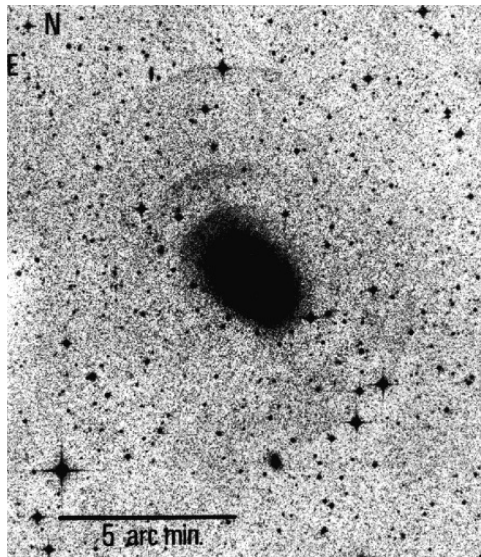


FIGURE 26.16 Faint concentric shells around the elliptical galaxy NGC 3923. (Figure from Malin and Carter, *Nature*, 285, 643, 1980. Anglo-Australian Observatory photo by David Malin.)

also reveal that 56% of all ellipticals have faint shells (as do 32% of all S0's). As shown in Fig. 26.16, otherwise normal ellipticals may have as many as twenty large concentric arcs of very low surface brightness, both outside and inside the galaxy. Numerical experiments find that these shells may be the result of head-on (or nearly head-on) collisions with smaller galaxies that have only a few percent of the mass of the larger elliptical. In essence, the captured stars slosh back and forth in the elliptical's gravitational well, like water rocking back and forth in a large bowl. Arcs are seen where the stars slow and reverse their courses. The natural spread in the stellar kinetic and potential energies creates the concentric arcs.

From the evidence discussed here and in Section 25.4, it seems likely that many, and perhaps all, large elliptical galaxies are strongly influenced by mergers. Galaxy mergers seem to produce results that are strikingly similar to elliptical galaxies in many ways. Photometry of the inner regions of cD galaxies and other obvious merger remnants confirms that their surface brightness profiles follow the $r^{1/4}$ law (Eq. 25.2) that is obeyed by most large ellipticals. Computer simulations bolster this view by showing that $r^{1/4}$ brightness profiles are normal outcomes of mergers. Calculations of mergers can also reproduce the observations of distinctly different velocity fields inside and outside the galactic core.

It is important to note, however, that dwarf ellipticals and dwarf spheroidals, the most common types of elliptical galaxies, do not appear to result from mergers.

Binary Supermassive Black Holes

An intriguing consequence of the merging of two large galaxies, whether they are ellipticals or spirals, is the apparently inevitable formation of a binary system of supermassive black holes. If each galaxy involved in the merger originally contained a supermassive black hole, the two black holes would migrate toward the center of the potential well of the combined system as a direct result of dynamical friction. Depending on their trajectories, the two black holes would probably enter into a well-separated binary system. One such system appears to be NGC 6420, which has two very strong X-ray sources (believed to be supermassive black holes) that are currently separated by a distance of about 1 kpc (see Fig. 26.17).

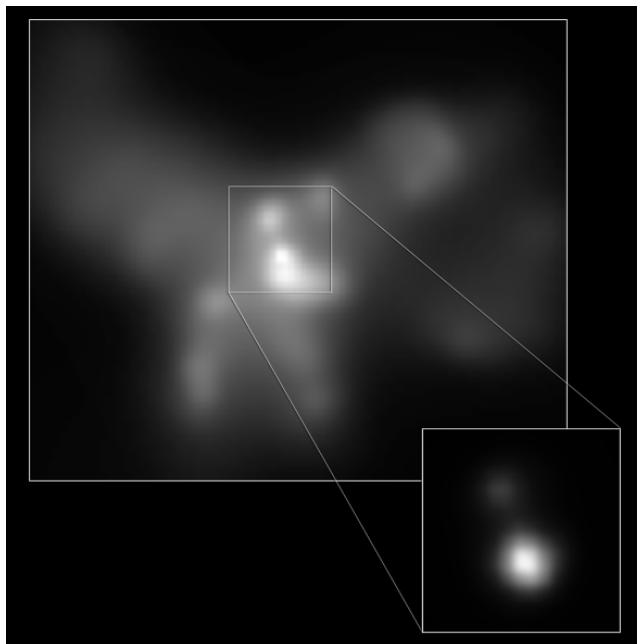


FIGURE 26.17 A Chandra X-Ray Observatory image of two supermassive black holes near the center of the “butterfly-shaped” galaxy, NGC 6420. (Courtesy of NASA/CXC/MPE/S. Komossa, et al.)

As the two supermassive black holes orbit around one another, they will have a profound and disturbing time-dependent gravitational influence on stars near the center of the merged galaxy. As with other three-body systems where one member is much less massive than the other two, it is probable that individual stars will be ejected from the central region of the galaxy. If the total mass of stars ejected in this way becomes comparable to the mass of the black holes, enough angular momentum and energy will be carried away by the stars that the two black holes will spiral in toward one another. When the black holes become sufficiently close, they will produce large amounts of gravitational radiation that will further rob them of the energy and angular momentum needed to stay separated (recall the discussion of gravitational radiation from binary pulsar systems in Section 18.6). The final outcome of the process will be a merger of the two black holes, producing an even larger beast in the center of the merged galaxy.

It has been suggested that such a mechanism is a probable route to growing the supermassive black holes that seem to be so prevalent in the centers of large galaxies.

26.2 ■ THE FORMATION OF GALAXIES

As we have seen so far in our discussions of the Milky Way and other galaxies, galactic structures are complex and varied. The luminous components of elliptical galaxies are dominated by spheroidal mass distributions that are primarily composed of old stars, while spiral galaxies contain both relatively old spheroids and appreciable disks of younger generations of stars, dust, and gas. Important differences even exist within the designations of early- and late-type galaxies, such as the degree of diskiness or boxiness of normal ellipticals, the existence of dwarf spheroidals, and the relative dominance of the bulges and disks of spirals, to name a few. Furthermore, the presence of large quantities of dark matter, accounting for 90% or more of the mass in many galaxies, also plays a critical role in determining their overall structures.

The Eggen, Lynden-Bell, Sandage Collapse Model

It was in 1962 that Olin J. Eggen (1919–1998), Donald Lynden-Bell, and Allan R. Sandage presented an important early attempt at modeling the evolution of our Galaxy, often referred to as the **ELS collapse model**. Their work was based on observed correlations between the metallicity of stars in the solar neighborhood, and their orbital eccentricities and orbital angular momenta. Eggen, Lynden-Bell, and Sandage noted that the most metal-poor stars tend to have the highest eccentricities, the largest w components of their peculiar motions (Section 24.3), and the lowest angular momenta about the rotation axis of the Galaxy. On the other hand, metal-rich stars tend to exist in nearly circular orbits and are confined to regions near the plane of the Galaxy.

To explain the kinematic and chemical properties of stars in the solar neighborhood, ELS suggested that the Milky Way Galaxy formed from the rapid collapse of a large protogalactic nebula. The oldest halo stars formed early in the collapse process while still on nearly radial trajectories, resulting in their highly elliptical orbits above and below the Galactic plane. As a further consequence of their rapid formation, the model predicts that the halo stars are naturally very metal-poor (Population II) since the interstellar medium had not

yet had time to become enriched by the by-products of stellar nucleosynthesis. However, as the first generations of massive stars generated heavier elements in their interiors, underwent supernova explosions, and ejected metal-rich material back into the ISM, the ISM evolved chemically over time.

As the proto-Galactic cloud continued to fall inward, the model predicts that the rapid collapse slowed when collisions between gas and dust particles became more frequent and the kinetic energy of infall was dissipated (converted into the thermal energy of random particle motions). Furthermore, the presence of angular momentum in the original proto-Galactic nebula meant that the cloud began to rotate more quickly as the radius decreased. The combination of the increased dissipation and the increased angular speed led to the development of a disk of chemically enriched gas from which Population I stars continue to form today.

Example 26.2.1. The time required for the free-fall collapse of the proto-Galactic cloud as envisioned by Eggen, Lynden-Bell, and Sandage can be estimated from Eq. (12.26), which we first used to study the collapse of a protostellar cloud. Assume that the proto-Galactic cloud contained some $5 \times 10^{11} M_{\odot}$, the estimated mass of the Milky Way Galaxy within a nearly spherical volume of radius 50 kpc (which includes the dark matter halo; see Table 24.1). If we further assume that the mass was uniformly distributed over the sphere, then the initial density of the cloud was

$$\rho_0 = \frac{3M}{4\pi r^3} = 8 \times 10^{-23} \text{ kg m}^{-3}.$$

Substituting into Eq. (12.26) gives

$$t_{\text{ff}} = \left(\frac{3\pi}{32} \frac{1}{G\rho_0} \right)^{1/2} = 200 \text{ Myr}.$$

Of course, if the nebula were initially somewhat centrally condensed, the inner portions of the Galaxy would collapse more rapidly than the outer, rarefied regions (see Problem 26.8). This may explain the existence of the very old stellar population within the bulge. The high metal abundance of bulge stars would arise if the first, massive, short-lived stars could quickly enrich the relatively dense ISM in that part of the Galaxy. Recall that the lifetimes of the most massive stars are on the order of one million years, much shorter than the estimated free-fall timescale.

Problems with the ELS Model

Although the model does account for many of the basic features found in the structure of the Milky Way, this *top-down* approach involving the differentiation of a single, immense proto-Galactic cloud does not explain several important aspects of our current understanding of the Galaxy's morphology. For instance, given an initial rotation of the proto-Galactic cloud, essentially all halo stars and globular clusters should be moving in the same general direction, albeit with highly eccentric orbits about the Galactic center. However, astronomers have come to realize that approximately one-half of all outer-halo stars are in retrograde

orbits (see Section 24.3) and the net rotational velocity of the outer halo is roughly 0 km s^{-1} . On the other hand, stars in the inner halo, along with the inner globular clusters, appear to have a small net rotational velocity. (Our current understanding of the kinematics of halo stars and clusters seems to suggest that the early environment of the Galaxy was fairly turbulent and clumpy.)

A second problem with the standard ELS collapse model is the apparent age spread among the globular clusters and halo stars. If the approximately 2-billion-year variation in ages discussed in Chapter 24 is real (the age range being perhaps 11 to 13 Gyr), then the collapse must have taken roughly an order of magnitude longer to complete than proposed by Eggen, Lynden-Bell, and Sandage. The model also does not readily explain the existence of a multicomponent disk having differing ages. Recall that the young disk is probably about 8 Gyr old, whereas the age of the thick disk may be 10 Gyr (both being significantly younger than the halo).

Yet another difficulty lies in the compositional variation found between globular clusters. The clusters located nearest the Galactic center are generally the most metal-rich and oldest, while the clusters in the outer halo exhibit a wider variation in metallicity and tend to be younger. The clusters also seem to form two spatial distributions; one set is associated with the spheroid, and the other may more properly be affiliated with the thick disk.

The problems that have developed with the early ELS view of the formation of the Milky Way suggest that our understanding of its formation and subsequent evolution must be revised or is otherwise incomplete. Furthermore, the rich variety of galaxies, along with their ongoing dynamical evolution via mutual interactions and mergers, poses interesting challenges to the development of an overall, coherent theory of galactic evolution. Although at the time this text was written, such a theory had not yet reached the same level of maturity that exists in our understanding of stellar evolution, important features have begun to emerge.

The Stellar Birthrate Function

As we have seen in the ELS model, any theory of galaxy formation must be able to explain the rate of formation of stars of various masses, as well as the chemical evolution of the interstellar medium. Since the ISM is enriched by mass loss via stellar winds and supernovae of various types, the theory must also incorporate the rates of stellar evolution and the chemical yields of stars.

One problem immediately arises in this regard: Although astronomers have been able to develop a reasonable description of the evolution of individual stars, our understanding of stellar birthrates is not yet complete. It is customary to express the **stellar birthrate function**, $B(M, t)$, in terms of the **star formation rate** (SFR), $\psi(t)$, and the **initial mass function** (IMF), $\xi(M)$, in the form

$$B(M, t) dM dt = \psi(t) \xi(M) dM dt, \quad (26.4)$$

where M is the stellar mass and t is time. $B(M, t)$ represents the number of stars per unit volume (or per unit surface area in the case of the Galactic disk) with masses between M and $M + dM$ that are formed out of the ISM during the time interval between t and $t + dt$. The SFR describes the rate per unit volume at which mass in the ISM is being

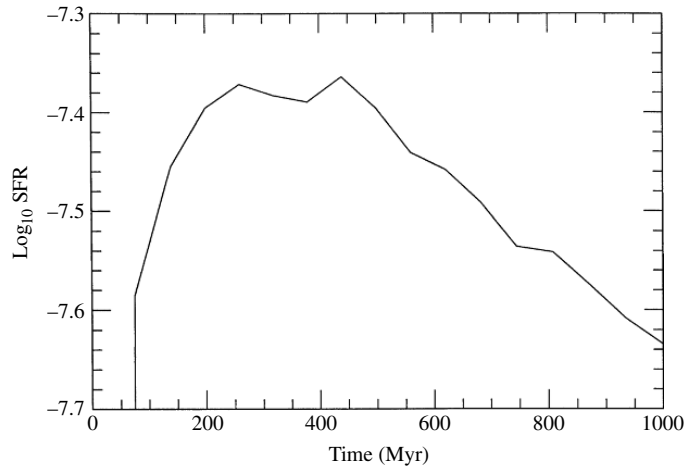


FIGURE 26.18 A model of the total star formation rate (in units of $M_{\odot} \text{ pc}^{-2} \text{ yr}^{-1}$) for the disk of the Milky Way Galaxy as a function of time. (Figure adapted from Burkert, Truran, and Hensler, *Ap. J.*, 391, 651, 1992.)

converted into stars; the present value for the SFR within the Galactic disk is believed to be $5.0 \pm 0.5 M_{\odot} \text{ pc}^{-2} \text{ Gyr}^{-1}$, integrated over the z direction (this corresponds to the two to three stars formed per year that was mentioned on page 1012). Finally, the IMF, which was first discussed in Section 12.3 (see Fig. 12.12), represents the relative numbers of stars that form in each mass interval.

To understand the birthrates of stars and their ensuing contribution to the chemical evolution of the ISM, different researchers have made various assumptions about the SFR. For instance, some astronomers have assumed that the SFR is time-independent, while others have argued for an exponentially decreasing function with time, or perhaps one that is proportional to some power of the surface mass density of the Galactic disk. One computer simulation of the evolution of the disk of our Galaxy, performed by Andreas Burkert, James W. Truran, and G. Hensler (1992), produced an SFR that reached a maximum value and then decreased with time as the available gas and dust in the ISM was consumed (see Fig. 26.18). Other studies have considered the possibility that the SFR may be highly variable in both space and time. Such a situation could occur because of short-timescale starburst activity, for instance.

The IMF is often modeled as a power-law fit of the form

$$\xi(M) = \frac{dN}{dM} = CM^{-(1+x)}, \quad (26.5)$$

where x may take on different values for various mass ranges and C is a normalization constant. The first attempt to derive an IMF for the solar neighborhood was carried out by Edwin E. Salpeter in 1955, where he argued for a value of $x = 1.35$. According to more recent determinations, it appears that $x = 1.8$ may be a better fit for stars in the approximate mass range $7 M_{\odot}$ to $35 M_{\odot}$ (and perhaps down as low as $2 M_{\odot}$). For stars more massive than about $40 M_{\odot}$, $x = 4$ may be required, implying that the production of massive stars drops

off very rapidly with increasing mass. For lower-mass stars, x is very difficult to determine through observational studies. The complications arise because the IMF must be decoupled from the **present-day mass function** and the SFR, which itself can be quite complicated. It has been suggested that the IMF may flatten considerably at low masses. Unfortunately, an exact form of the IMF is not yet known, and it is not even clear whether the IMF varies with time or location.

The G-Dwarf Problem

Attempts to use the SFR and IMF to model the chemical evolution of the Galactic disk in the solar neighborhood have resulted in predictions that do not always agree with observations. If we assume that the first generation of stars was born without any metals ($Z_0 = 0$; Population III) and the chemical evolution of the ISM occurs within a *closed box* (meaning that no gas or dust is allowed to enter or leave the system being modeled), then the calculations predict too many stars of low metallicity when compared with observations. For instance, this simple model suggests that roughly one-half of the stars in the solar neighborhood should have values for Z that are less than one-quarter of the solar value ($Z_\odot \simeq 0.02$). However, only about 2% of the F and G main-sequence stars have such low Z values. This is known as the **G-dwarf problem**.

One possible solution to the G-dwarf problem, referred to as **prompt initial enhancement**, is to assume that the disk of our Galaxy formed with $Z_0 \neq 0$, which could occur if heavy-element enrichment of the ISM resulted from rapidly evolving massive stars before the gas and dust settled into the disk. A second suggestion is that the disk accumulated mass over a significant period of time, perhaps even continuing to the present (in other words, the closed-box assumption is invalid). In this scenario a substantial infall of metal-poor material onto the Galactic disk has occurred since its initial formation; as the gas entered the system, it mixed with the metal-enriched ISM. Since a lower initial mass density would imply fewer stars formed during the early history of the disk, fewer metal-poor stars would be observed today. Yet another proposal argues that the IMF was different in the early history of the Galaxy, and a larger fraction of more massive stars were formed with correspondingly fewer low-mass stars. Since massive stars are short-lived, this hypothesis would result in fewer metal-poor stars today.

A Dissipative Collapse Model

Another issue that must be addressed by a comprehensive theory of galactic evolution is the question of a free-fall collapse versus a slow, dissipative one. A free-fall collapse is governed by the free-fall timescale, sometimes alternatively referred to as the **dynamical timescale**, given by Eq. (12.26) (see Example 26.2.1). On the other hand, a dissipative collapse can be described in terms of the time necessary for the nebula to cool significantly. If the **cooling timescale**, t_{cool} , is much less than the free-fall timescale, then the cloud will not be pressure-supported and the collapse will be rapid (i.e., essentially in free-fall). However, if the cooling time exceeds the free-fall timescale, the gas cannot radiate its energy away fast enough to allow for a rapid collapse, and the gravitational potential energy that is released during the collapse will heat the nebula adiabatically. This situation is yet another example of the virial theorem.

To estimate the cooling timescale, we must first determine the characteristic amount of thermal kinetic energy contained within each particle in the gas. According to the virial theorem, if we assume that the gas is in quasistatic equilibrium, the average thermal kinetic energy of the gas must be related to its potential energy by Eq. (2.46),

$$-2 \langle K \rangle = \langle U \rangle.$$

If we further assume that the gas has a mean molecular weight of μ and contains N particles, then the virial theorem gives

$$-2N \frac{1}{2} \mu m_H \langle v^2 \rangle = -\frac{3}{5} \frac{GM^2}{R},$$

where $m = \mu m_H$ is the average mass of a single particle, R is the radius of the nebula, and $M = N \mu m_H$ is the nebula's mass. In the last expression, we used a gravitationally bound, spherical mass distribution of constant density to estimate the potential energy of the system; see Eq. (10.22). Solving for the velocity dispersion $\sigma = \langle v^2 \rangle^{1/2}$ gives

$$\sigma = \left(\frac{3}{5} \frac{GM}{R} \right)^{1/2}. \quad (26.6)$$

Note the similarity of this expression to Eq. (25.13) for the virial mass of a spherical cluster of stars.

Now we may determine a characteristic temperature of the gas, known as the **virial temperature**, by equating the typical kinetic energy of a gas particle to its thermal energy, or

$$\frac{1}{2} \mu m_H \sigma^2 = \frac{3}{2} k T_{\text{virial}},$$

which gives

$$T_{\text{virial}} = \frac{\mu m_H \sigma^2}{3k}. \quad (26.7)$$

Finally, to estimate the cooling time we must determine the rate at which energy can be radiated away from the gas. This is done by expressing the cooling rate per unit volume as

$$r_{\text{cool}} = n^2 \Lambda(T),$$

where r_{cool} has units of energy per unit time per unit volume, n is the number density of particles in the gas, and $\Lambda(T)$ is a quantum mechanical **cooling function**. $\Lambda(T)$, shown in Fig. 26.19, includes the same physical processes that were discussed in the context of stellar opacity in Section 9.2 (bound-bound, bound-free, free-free, and electron scattering). The two “bumps” in the curve just above 10^4 K and near 10^5 K correspond to the ionization/recombination temperatures of hydrogen and helium, respectively. Above about 10^6 K, the cooling is due to thermal bremsstrahlung and Compton scattering. The n^2 dependence in the expression for r_{cool} can be understood in terms of the interactions between

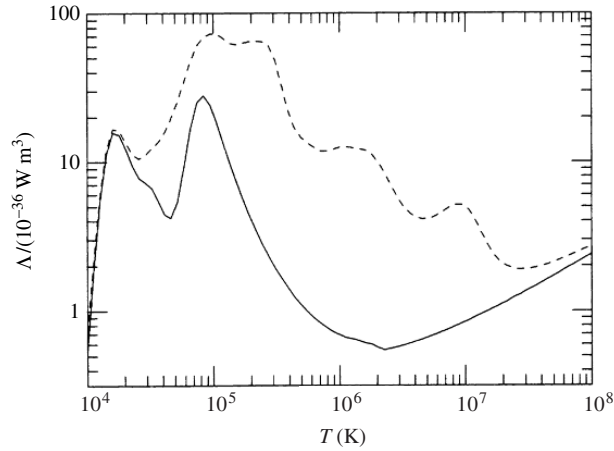


FIGURE 26.19 The cooling function $\Lambda(T)$. The solid line corresponds to a gas mixture of 90% hydrogen and 10% helium, by number. The dashed line is for solar abundances. (Figure adapted from Binney and Tremaine, *Galactic Dynamics*, Princeton University Press, Princeton, NJ, 1987.)

pairs of particles in the gas; collisions excite ions, atoms, or molecules, which then radiate the energy away in the form of photons, cooling the gas.

If all of the energy in the cloud is radiated away in a time t_{cool} , then

$$r_{\text{cool}} V t_{\text{cool}} = \frac{3}{2} N k T_{\text{virial}},$$

where V is the volume of the cloud. Solving for the cooling time, we have

$$t_{\text{cool}} = \frac{3}{2} \frac{k T_{\text{virial}}}{n \Lambda}. \quad (26.8)$$

Example 26.2.2. Assume for simplicity that the proto-Galactic nebula discussed in Example 26.2.1 was initially composed of 90% hydrogen and 10% helium, by number (this corresponds to $X \simeq 0.7$ and $Y \simeq 0.3$).⁵ If we assume complete ionization, then the mean molecular weight is given by Eq. (10.16), or $\mu \simeq 0.6$. Also, according to Eq. (26.6), the initial velocity dispersion of the particles in the gas was approximately $\sigma = 160 \text{ km s}^{-1}$. Then, substituting into Eq. (26.7), the virial temperature of the gas at the time of collapse was roughly

$$T_{\text{virial}} \simeq 6 \times 10^5 \text{ K}.$$

(At this temperature our assumption that the gas was completely ionized is certainly valid.)

⁵Note that we are assuming here that *all* of the mass is in the form of baryonic matter. This means that we are neglecting the influence of dark matter in this example.

The number density of particles in the gas is given by

$$n = \frac{\rho}{\mu m_H} = \frac{3M}{4\pi R^3 \mu m_H} \sim 5 \times 10^4 \text{ m}^{-3}.$$

This value should be compared to the typical number densities found in Galactic molecular clouds today, which are on the order of $n \sim 10^8$ to 10^9 m^{-3} (see Section 12.1).

Now, from Fig. 26.19, $\Lambda \sim 10^{-36} \text{ W m}^3$ and the cooling time for the cloud (Eq. 26.8) is found to be

$$t_{\text{cool}} = 8 \text{ Myr}.$$

Clearly, in this case $t_{\text{cool}} \ll t_{\text{ff}}$. Apparently the proto-Galactic nebula was capable of radiating energy away at a rate sufficient to allow for a free-fall collapse.

It is instructive to consider the situation where $t_{\text{cool}} > t_{\text{ff}}$. In this case the nebula is unable to efficiently radiate away the gravitational potential energy that is released by the collapse. As a result, the cloud's temperature would rise adiabatically as the cloud shrinks, resulting in an increasing internal pressure and a halt to the collapse. After the collapse has halted, the virial theorem governs the equilibrium conditions of the cloud. For the approximate values of $T \sim 10^6 \text{ K}$ and $n \sim 5 \times 10^4 \text{ m}^{-3}$ characteristic of a protogalactic cloud at the time of formation of the first galaxies, the upper limit on the mass that can cool and collapse is on the order of $10^{12} M_{\odot}$, with a corresponding radius of about 60 kpc. In regions of the cloud where the gas temperature had decreased to the level of hydrogen recombination, $T \sim 10^4 \text{ K}$, the mass limit becomes $\sim 10^8 M_{\odot}$. Thus the galaxies that are observed today would be expected to have masses in the range from $10^8 M_{\odot}$ to $10^{12} M_{\odot}$. The lower limit corresponds fairly well with the values of the smallest dwarf elliptical galaxies, and the upper limit agrees with the values measured for the most massive giant spiral galaxies (Sa–Sc). Although some giant ellipticals and cD galaxies exceed $10^{12} M_{\odot}$, they have certainly been affected significantly by mergers throughout their histories and are not in virial equilibrium near their outer edges.

Although the proto-Galactic cloud was able to radiate away the initial release of gravitational energy content from the system, shortly after the collapse began a new source of energy became available. The deaths of the first generations of very massive stars meant that supernova shock waves moved through the ISM at speeds on the order of $0.1c$. As the expanding shells struck the gas, the gas was reheated to temperatures of a few million kelvins, slowing the rate of collapse somewhat. However, calculations of the supernova production rate that are based on estimates of the IMF seem to indicate that even this new source of energy was unable to slow the collapse appreciably.

The Hierarchical Merger Model

How then can we explain the apparent age and metallicity differences among the globular clusters, as well as the existence of distinct components of varying ages within the disk of our Galaxy (i.e., the thick disk and the younger thin disk)? The answer appears to be that not only did galaxy-building involve the top-down process originally envisioned by Eggen, Lynden-Bell, and Sandage, but it also incorporated a *bottom-up hierarchical process of mergers*.

With the realization in the 1970s and 1980s that mergers play an important role in galactic evolution, and because of observational and theoretical developments regarding the nature of the early universe, the hierarchical merger scenario has received a great deal of attention. Scientists now believe that shortly after the birth of the universe via the Big Bang, density fluctuations existed in the overall distribution of matter. (The details of the evolution of the universe as a whole will be discussed in Chapter 30.) Our current understanding of those fluctuations suggests that the most common density perturbations occurred on the smallest mass scales. Consequently, density fluctuations involving 10^6 to $10^8 M_\odot$ were much more common than those for $10^{12} M_\odot$ or more.

First consider the formation of the Milky Way as an example of the hierarchical merger process. As these 10^6 to $10^8 M_\odot$ proto-Galactic fragments were gravitationally attracted to one another, they began to merge into a growing spheroidal mass distribution. Initially, many of the fragments evolved in virtual isolation, forming stars and, in some cases, globular clusters in their centers. As a result, they developed their own chemical histories and unique abundance signatures. In the inner regions of the growing spheroid, where the density of matter was greater, its rate of collapse and subsequent evolution would have been more rapid. This resulted in the production of the oldest stars that are observed today, together with a greater degree of chemical enrichment (hence the old, metal-rich central bulge). In the rarefied outer regions of the Galaxy, chemical evolution and star formation would have been much slower.

According to the hierarchical model, collisions and tidal interactions between merging fragments disrupted the majority of the fragments and left exposed the globular cluster cores of others. Furthermore, in this model the disrupted systems would have led to the present distribution of the field halo stars, while leaving the remaining globular clusters scattered throughout the spheroid. Those proto-Galactic fragments that were initially moving in a retrograde direction relative to the eventual orbital motion of the Galactic disk and inner halo produced the net zero rotation of the outer halo that is observed today.

Certainly the rate of collisions would have been greater near the center of the Galaxy, disrupting those proto-Galactic fragments first and building the bulge more rapidly than the halo. Consequently, according to this picture, the spheroidal component of the Galaxy can be considered as forming from the inside out.

The globular clusters still present in the Galaxy today probably total only some 10% of the number that originally formed from proto-Galactic fragments. The other 90% were disrupted by collisions and tidal interactions during the early merger process and by the subsequent ongoing effects of dynamical friction. This may help to explain the relative uniformity in the masses of globular clusters observed today (approximately 10^5 to $10^6 M_\odot$). Low-mass globular clusters would have had small gravitational binding energies, allowing them to be disrupted comparatively easily and rapidly when the Galaxy was young. On the other hand, recall from Eq. (26.1) that dynamical friction is strongly dependent on the mass of the cluster ($f_d \propto M^2$) so that massive clusters would have spiraled rapidly into the inner regions of the Galaxy where stronger and more frequent interactions ultimately disrupted them as well.

It is important to note that because of their isolation in the outer reaches of the Galaxy and the slower rate of evolution there, the proto-Galactic fragments in that region would have evolved almost like individual dwarf galaxies for a time. In fact, the significant number of dSph galaxies still present in the Local Group are assumed to be surviving proto-Galactic

fragments. In addition, there is clear evidence of ongoing mergers today, such as the dwarf spheroidal Sagittarius galaxy and the Magellanic Stream, indicating that the construction of the Milky Way's halo is still a work in progress.

Formation of the Thick Disk

As the gas clouds of disrupted proto-Galactic fragments collided, the collapse became largely dissipative. This means that the gas began to settle slowly toward the central regions of the Galaxy. However, because of the presence of some initial angular momentum in the system, introduced through torques from other neighboring protogalactic clouds, the collapsing material eventually became rotationally supported and settled into a disk about the Galactic center. Of course, the already-formed halo stars did not participate in the collapse to the disk, because their collisional cross sections were now too small to allow them to interact appreciably, except through gravitational forces.

One model of thick-disk formation suggests that the thick disk may have formed around the Galactic midplane with a characteristic temperature of $T \sim 10^6$ K. By equating the kinetic energy of a typical particle in the gas to its gravitational potential energy above the midplane of the disk, the approximate scale height, h , of the disk of gas can be estimated.

To determine the local acceleration of gravity, g , at a height, z , above the midplane, imagine that the disk has a mass density, ρ , given by

$$\rho(z) = \rho_0 e^{-z/h},$$

where ρ_0 is the mass density in the Galactic midplane (see Eq. 24.9). Now, according to the gravitational version of “Gauss’s law” [see Eq. (24.56) in Problem 24.26],

$$\oint \mathbf{g} \cdot d\mathbf{A} = -4\pi G M_{\text{in}},$$

where the integral is over a closed surface that bounds the mass M_{in} , and $\mathbf{g}$ is the local acceleration of gravity at the position of $d\mathbf{A}$. If h is much smaller than the diameter of the disk, then for a Gaussian cylinder of height $2h$ and cross-sectional area A , centered on the midplane (see Fig. 26.20),

$$2Ag = 4\pi G M_{\text{in}},$$

where M_{in} is the amount of disk mass contained within the cylinder. M_{in} can be estimated by integrating the mass density throughout the volume of the cylinder, or

$$M_{\text{in}} = 2 \int_0^h \rho_0 e^{-z/h} A dz = 1.26 \rho_0 A h.$$

Substituting into our previous expression, we have that the local acceleration of gravity at a height, h , is given by

$$g(h) = 2.53\pi G \rho_0 h.$$

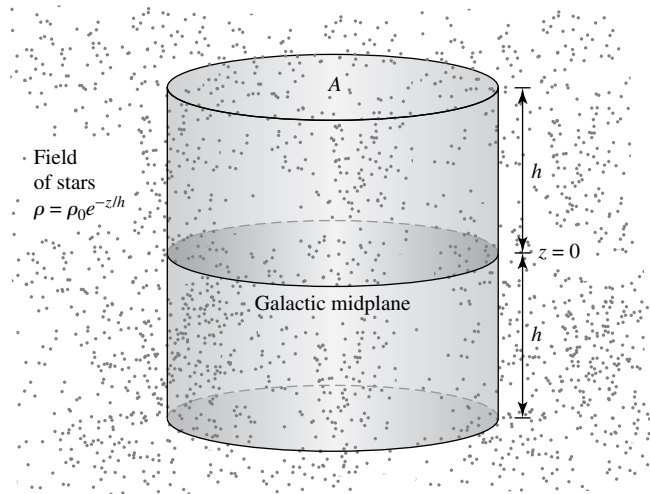


FIGURE 26.20 A Gaussian cylinder located entirely within the field of stars of the Galactic disk.

Next, the gravitational potential energy of a particle of mass, m , at a height, h , above the midplane is given by

$$U(h) = \int_0^h mg(z) dz = 1.26\pi Gm\rho_0 h^2.$$

Equating the potential energy to the average thermal kinetic energy of a particle, $K = 3kT/2$, we find that

$$h(T) = \left(\frac{3kT}{2.53\pi Gm\rho_0} \right)^{1/2}. \quad (26.9)$$

Example 26.2.3. If the gas in the proto-Galactic thick disk had a characteristic temperature of $T \sim 10^6$ K, and if we assume that the central mass density was comparable to the value that is estimated today for the solar neighborhood,

$$\rho_0 \simeq 0.15 \text{ M}_\odot \text{ pc}^{-3} = 1.0 \times 10^{-20} \text{ kg m}^{-3},$$

then

$$h(10^6 \text{ K}) \simeq 2.2 \text{ kpc},$$

where we have used the mass of a hydrogen atom for m . From Table 24.1, the measured value for the scale height of the thick disk is approximately 1 kpc.

In regions where the gas was locally more dense, it cooled more rapidly, since $t_{\text{cool}} \propto n^{-1}$ (see Eq. 26.8). This was accomplished first through thermal bremsstrahlung and Compton scattering, and then, when the temperature reached $\sim 10^4$ K, via the radiation emitted

by hydrogen atoms. This meant that once the hydrogen recombination temperature was reached, H I clouds could form and begin producing stars. Within a few million years, the most massive stars underwent core-collapse supernova detonations and their shocks began to reheat the gas between the molecular clouds, maintaining the temperature of the intercloud gas at roughly 10^6 K. At the same time, the production of iron in the supernovae raised the metallicity from an initial value of $[\text{Fe}/\text{H}] < -5.4$ (perhaps originally enriched somewhat by the stars that formed in the proto-Galactic fragments) to $[\text{Fe}/\text{H}] = -0.5$. About 400 million years after the first stars were created in the thick disk, star formation nearly ceased. In total, a few percent of the mass of the gas was converted into stars during this thick-disk-producing period of the Galaxy's evolution.

A modified version of the thick-disk formation model described above suggests that the infalling gas was initially much cooler. This meant that the gas (and dust) was able to settle onto the midplane with a much smaller scale height, similar to today's thin disk. Star formation was then able to proceed due to the greater local density of gas and dust. However, as a direct result of a significant merger event with a proto-Galactic fragment some 10 Gyr ago, the disk was reheated by the energy of the interaction, causing it to puff up to its present 1-kpc scale height.

Formation of the Thin Disk

After the formation of the thick disk, cool molecular gas continued to settle onto the midplane with a scale height of approximately 600 pc. During the next several billion years, star formation occurred in the thin disk.

The process of maintaining the scale height was essentially a self-regulating one. If the disk became thinner, its mass density would increase. This in turn would cause the SFR to increase, producing more supernovae and reheating the disk's intercloud gas component. The ensuing expansion of the disk would again decrease the SFR, yielding fewer supernovae, and the disk would cool and shrink. However, despite the self-regulating process, as the gas was depleted in the ISM the SFR decreased from about $0.04 \text{ M}_\odot \text{ pc}^{-3} \text{ Myr}^{-1}$ to $0.004 \text{ M}_\odot \text{ pc}^{-3} \text{ Myr}^{-1}$. At the same time, the metallicity continued to rise, reaching a value of approximately $[\text{Fe}/\text{H}] = 0.3$. Because of the decrease in the SFR, the thickness of the disk decreased to about 350 pc, the scale height of today's thin disk. During the development of the thin disk, some 80% of the available gas was consumed in the form of stars.

Finally, as the remaining gas continued to cool, it settled into an inner, metal-rich and gas-rich component of the thin disk with a scale height of less than 100 pc. Today most ongoing star formation occurs in this young, inner portion of the thin disk, the component in which the Sun resides.

The Existence of Young Stars in the Central Bulge

The existence of young stars in the central bulge of our Galaxy can be understood in the context of the evolution just described by arguing for recent mergers with gas-rich satellite galaxies. When those galaxies were disrupted by tidal interactions with the Milky Way, their gas settled into the disk and the center of the Galaxy, ultimately forming new stars. It also appears that the Milky Way's central bar plays a role in the migration of dust and gas into the inner portion of the Galaxy by generating dynamical instabilities as it rotates.

Metallicity Gradients

The hierarchical merger scenario just outlined predicts that metallicity gradients ought to exist in galaxies that have undergone a dissipative collapse. If a galaxy is more metal-rich in its center than it is near the outskirts of the system, then a color gradient should also exist. Because of the enhancement of opacity with metallicity (see Section 25.2), the galaxy would be redder in its center than it is farther out.

Of course, the strength of the metallicity and color gradients can be diminished or even destroyed by sufficiently frequent and energetic mergers with other galaxies. For instance, many starburst galaxies actually have inverted color gradients and appear bluer in their centers. This is because of the large SFR that resulted from the sudden influx of gas-rich material in the galactic center when another galaxy was disrupted, or from the effects of tidal torques that acted on the starburst galaxy itself, causing its own gas to spiral into the center. It is worth noting that weaker metallicity gradients are observed in boxy ellipticals than in disk galaxies, another possible indication that boxy E's have undergone significant merger activity during their lifetimes.

The Formation of Elliptical Galaxies

Although we do not yet understand all of the complex details of galactic evolution, the nature of the Hubble sequence may ultimately correspond to the mass of the individual galaxies, the efficiency with which the galaxies made stars, and the relative importance of free-fall collapse, dissipative collapse, and mergers. As an example, it appears that many ellipticals may have formed the majority of their stars early in the galaxy-building process, before the gas had a chance to settle into a disk, whereas late-type galaxies took a more leisurely pace. You may also recall that current observations indicate that later Hubble types have a higher relative abundance of gas and dust in their disks than galaxies of earlier Hubble type (see Tables 25.1 and 25.2).

However, as first mentioned at the end of Section 26.1, other E's probably formed from the collisions of already-existing spirals. The energy involved in the collision would destroy the disks of both galaxies and cause the merged system to relax to the characteristic $r^{1/4}$ distribution of an elliptical.

Although N -body simulations have been able to produce such a result (see Fig. 26.21), some questions remain. For instance, the large number of globular clusters in E's relative to spirals present a serious difficulty in arguing that cataclysmic collisions are the cause of all large elliptical galaxies [Fig. 25.16 and Eq. (25.15)]. On the other hand, perhaps mergers can actually produce globular clusters by triggering star formation in clouds, in which case the larger specific frequency of globular clusters may not be a problem. It is also possible that many of the observed globular clusters are captured dwarf spheroidal galaxies, just as ω Cen appears to be in the Milky Way.

From the discussion at the beginning of Section 26.1, recall that elliptical galaxies are much more abundant relative to spirals in the centers of dense, rich clusters of galaxies, whereas spirals dominate in less dense clusters and near the periphery of rich clusters. (This **morphology–density relation** was first reported by Alan Dressler in 1980.) This effect may be partly explained by the increased likelihood of interactions in regions where galaxies are more tightly packed, destroying spirals and forming ellipticals.

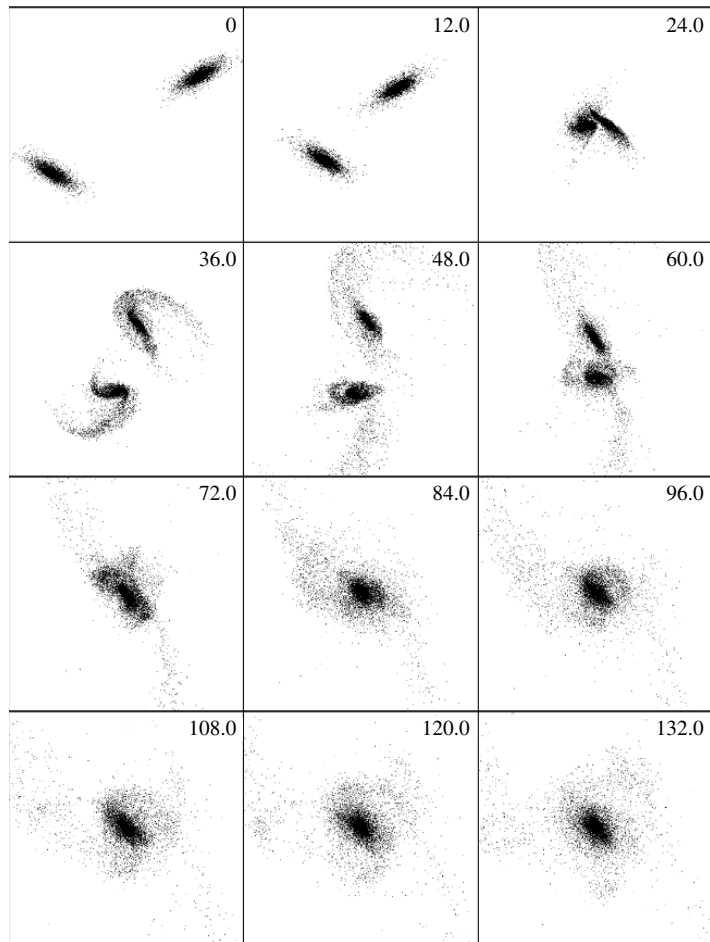


FIGURE 26.21 An N -body simulation of the merger of two spiral galaxies. Each disk is represented by 16,384 particles, and each bulge contains 4096 particles. The result is an elliptical galaxy with an $r^{1/4}$ profile. (Figure adapted from Hernquist, *Ap. J.*, 409, 548, 1993.)

However, a competing hypothesis has also been suggested—namely, that ellipticals tend to develop preferentially near the bottoms of deep gravitational potential wells, even in the absence of interactions. Lower mass-density fluctuations in the early universe may have resulted in spiral galaxies, and the smallest fluctuations led to the formation of dwarf systems. If this is the case, then the large number of dSph's and dE's that exist has a natural explanation in the much larger number of smaller fluctuations that formed in the early universe. This mechanism could help explain galactic morphology if the initial density fluctuations in the early universe were largest in what later became the centers of rich clusters. Because the gravitational potential well in those regions would have been deeper, the probability of collisions between protogalactic clouds should have been correspondingly greater as well.

Perhaps the frequency of collisions between protogalactic clouds is correlated with galaxy morphology.

Galaxy Formation in the Early Universe

Given the large number of fundamental problems that still remain in our development of a coherent theory of galaxy formation, it is fortunate that a means exists (at least in principle) for testing our ideas by observing galactic evolution through time. Because of the finite speed of light, when astronomers look farther and farther out into space, we are literally looking farther and farther back in time. For instance, galaxies that are 1 Mpc away emitted their light more than 3 million years ago. This means that the light we observe today coming from the galaxy is itself more than 3 million years old; we are “seeing” the galaxy 3 million years in the past.

In 1978, Harvey Butcher and Augustus Oemler, Jr., noted that there appeared to be an overabundance of blue galaxies in two distant clusters. They speculated that there may have been a significant evolution in galaxies over time, leading to the types of objects we see closer to us today. The **Butcher–Oemler effect**, as it is now called, suggests that galaxies in the early universe were bluer on average than they are today, indicating an increased level of star formation. The effect has been confirmed in numerous recent studies.

The morphology–density relation has also been shown to be time-dependent. As observations probe earlier times (more distant galaxy clusters), elliptical, and lenticular galaxies become less abundant relative to spiral galaxies, suggesting an evolution from later Hubble types to earlier types over time. This is just what would be expected if some earlier Hubble-type galaxies form from the mergers of spirals. This is also consistent with the overall picture of hierarchical galaxy building.

In 2004, the Space Telescope Science Institute released the Hubble Ultra Deep Field (HUDF) image shown in Fig. 26.22. The HUDF image is actually a composite of two images, one taken by the Advanced Camera for Surveys (ACS) and the other from the Near-Infrared Camera and Multi-Object Spectrometer (NICMOS). The region of the sky contained in the HUDF image is only 3 arcmin on a side (about 1/10 the size of the full moon), centered on ($\alpha = 3^{\text{h}}32^{\text{m}}40.0^{\text{s}}$, $\delta = -27^{\circ}48'00''$) in the constellation of Fornax. The image required a total exposure time of 11.3 days for ACS (obtained between September 24, 2003, and January 16, 2004) and 4.5 days for NICMOS (over the period between September 3, 2003, and November 27, 2003).

The HUDF reveals some very distant galaxies as they existed just 400 Myr to 800 Myr after the Big Bang. As is evident in the close-up of a portion of the HUDF shown in Fig. 26.23, the very distant galaxies appear quite different from the relatively nearby majestic spirals and ellipticals that are seen in the present-day universe.

Observations like these suggest that the abundance of strange-looking, remote, blue galaxies seen in the early universe may be the building blocks of today’s Hubble sequence of galaxies. They probably represent the proto-galactic fragments responsible for the hierarchical mergers that are still occurring today at a much diminished rate.

In Section 30.2 we will turn our attention to the earliest formation of structure in the universe, including the clumping of dark matter that likely formed the potential wells in which the clusters of galaxies and their diverse membership evolved. Certainly a great deal